UNITED STATES DEPARTMENT OF THE INTERIOR

NATIONAL PARK SERVICE

PACIFIC WEST REGION

YOSEMITE NATIONAL PARK

REPORT OF WASTE DISCHARGE

GLACIER POINT RESTROOM WASTEWATER EFFLUENT DISPOSAL SYSTEM

Prepared by Fulce Engineering, Inc.

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REGISTERED PROFESSIONAL ENGINEER
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Section 1: Background

Section 1.1: Wastewater System Description

The Report of Waste Discharge (ROWD) for the Glacier Point Restroom Wastewater Effluent Disposal System presents the wastewater facilities and effluent disposal facilities serving the Glacier Point Area of Yosemite National Park. The report provides the information requested by the California State Water Resources Control Board as it relates to Order WQ 2014-0153-DWQ: General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems, dated September 23, 2014.

Glacier Point is located in Yosemite National Park at the terminus of Glacier Point Road, approximately 16 miles east of Highway 41, at an elevation of 7,200 feet above Mean Sea Level. The Glacier Point wastewater system is in Township 2S, Range 22E, MDB&M, in which the surface water drainage for the area generally follows natural topography to the east and collects in Illilouette Creek. The site is located within the Yosemite hydrologic area. The wastewater system serves visitors who come to this location for site seeing and outdoor activities, along with employees who work at the visitor facilities (See Figure 1 for Location and Vicinity Map).

Wastewater is generated at three general locations:

> Area 1: Existing Visitor Comfort Station, which includes a six water closets and three sinks on the women's side and three water closets, three urinals and two sinks on the men's side.
>
> Area 2: Existing Day Use Concessions Building, which includes a kitchen area and a single employee restroom with one water closet and one sink.
>
> Area 3: Existing Seasonal National Park Service (NPS) Employee Living Quarters – this consists of five (5) small cabins that house employees from approximately May through October. Each cabin has its own dish sink. In addition, there is a common toilet facility and shower unit for the residences.

(See Figure 2 for Site Map)

The wastewater system operates as follows:

- Wastewater for Area 1 and 2 are collected by gravity at a 15,000 gallon septic tank, where effluent is pumped through a lift station approximately 2,000 feet (elevation gain of 240 feet) to a new concrete dosing tank (485 gallon dual siphon). Elevation at this collection lift station is approximately 7,190 feet.
- Wastewater for Area 3 is collected at a new lift station adjacent to the employee cabins and pumped approximately 460 feet (elevation gain of 65 feet) to a new 2,000 gallon septic tank. From the 2,000 gallon septic tank, effluent is gravity fed to the new concrete dosing tank, where it is co-mingled with the effluent from the Visitor Comfort Station and Concessions Building. Elevation at this collection system is approximately 7,365 feet.
- From the Dosing Tank, effluent is gravity fed to a splitter box, where it is transferred to four leachfield zones (A – D). Elevation at this system is approximately 7,430 feet.

- The leachfields have a total length of 3,141 linear feet and a capacity is 15,000 gallons per day. The leachfield is located on the natural topography at a slope from southwest to northeast at approximately 19%. Elevation of leachfield is approximately 7,400 to 7,420 feet.

(See Figure 3 for the Wastewater System Flow Diagram)

Domestic water to the Glacier Point area is served by a radial well located at Pothole Meadows 2,650 feet to the south of the leachfield, as well as a secondary surface water source at Union Point approximately 6,000 feet northwest of the leachfield area Water is treated at a treatment shed next to a 311,000 gallon storage tank located approximately 1,000 feet from the leachfields. All domestic water services (Glacier Point visitor area and the Employee Quarter's Cabins) are fed by gravity. There are no surface water bodies in the area. The water system serves the Visitor Area Comfort Station and Concessions Building through a 6” gravity water line, which is approximately 1,300 feet from the water storage tank. The employee area is served by a 3” HDPE gravity waterline and is located approximately 1,200 feet from the water storage tank.

Section 1.2: Service Area Description

The Glacier Point Wastewater system is an isolated and self-contained system, not connected to a regional wastewater collection system. Wastewater is collected and effluent disposed in the specific area. Septic tanks are pumped at the end of the visitor season (typically October) and waste disposed of at the El Portal Wastewater Treatment Plant. The El Portal Wastewater Treatment Plant is operated by the National Park Service and located in El Portal, CA at the west side of Yosemite National Park. It is located 18 miles from Glacier Point at an elevation difference of 5,400 feet. It is not feasible to connect the Glacier Point system directly to the El Portal Wastewater Plant.

The wastewater collection system details are as follows:

- The visitor area collection system was constructed in 1997.
- The Employee Quarters area collection system was originally constructed in the mid-1900’s. The lift station serving the area was constructed in 2019.
- The lift stations both have redundant pumps and a local warning alarm/light to indicate if high water level is reached in the lift station. When the system is in operation (approximately May through October), the system is checked daily by NPS maintenance personnel.
- NPS Maintenance personnel monitor the effluent disposal system daily when it is in operation. Spill response is provided by maintenance personnel and includes standard procedures to close the waste generation facilities and provide containment and cleanup upon any spill or leak.
- The effluent disposal system, including the leachfield is located south of the Employee Quarters was constructed in 2019.

The storm water collection system consists of natural surface runoff, which flows generally east along the natural topography through ephemeral surface water drainage channel tributaries to Illilouette Creek. Illilouette Creek is a tributary to the Merced River. There are no formal collection systems for storm water runoff. The leachfield site is generally graded away from the storm water runoff, such that there is only a minor amount of storm water that may infiltrate into the leachfield system.

Section 2: Wastewater Characterization and Treatment

Section 2.1: Domestic Wastewater Characterization

The generation of the untreated wastewater is from the Visitor Comfort Station primarily, which constitutes the majority of wastewater for the system. This is typical human waste from water closets and sinks. Additional wastewater flows are generated from the single water closet and sink at the Concessions Building, as well as a small amount of kitchen wastewater that consists of organics and detergents, such as kitchen serving utensil. Wastewater generated from the employee living quarters is consistent with residential waste generation. RV waste disposal is not allowed at Glacier Point.

Domestic wastewater flow rates were calculated based on estimated historical usage during the peak visitation season to Glacier Point. Historical flows, with safety factors, are 14,243 gallons per day at the Visitor's area. Calculated flows for the Employee Quarters are 627 gallons per day based on fixture counts. The system was sized for a total flowrate of 15,000 gallons per day. See attached Figure 4 – Design Criteria for flow and volume calculations summary. In addition, see Appendix A for Design Calculations.

Section 2.2: Wastewater Treatment System

The wastewater treatment system consists of septic tanks, dosing tanks and leachfields to dispose of wastewater at Glacier Point. See Figure 3 for the Wastewater System Flow Diagram.

The septic tanks (2) are pumped as needed during the visitor season and a final pumping occurs at the end of the season once the facilities are shut down for the winter (typically October). The lift stations are also pumped out at the end of the season when the system is winterized. Pretreatment waste is pumped by a contracted waste collection company and disposed of at the El Portal Wastewater Treatment Plant in El Portal, California.

Since this is a septic/leachfield system, there are no preliminary treatment or primary treatment processes/technology associated with the system, nor is there disinfection included in the waste disposal system.

The treated effluent disposal method is a leachfield system constructed in 2019. The leachfield consists of four zones with 7 laterals each in three zones and 9 laterals in one zone. The lines utilize a distribution box at each lateral which distributes effluent each direction. Once the distribution box fills up, it overflows to the downstream distribution box/lateral. The system will typically operate one zone at a time with daily review of monitoring wells which will indicate to maintenance when to rotate effluent to the next zone. If needed, multiple zones can be utilized at the same time. Twelve monitoring wells are located in and around the 0.68 acre leachfield site. See Figure 2 for Site Plan of leachfield area and Figure 5 for the Leachfield Distribution System Profile.

The surrounding land use is forested area. Drainage ways are located around the leachfield site. Depth to groundwater varies depending on season. There are no surface waters in the area and the nearest well is 2,650 feet to the south.

The percolation rates sampled prior to construction of the leachfield area are summarized in the below table:

TABLE 1
SUMMARY OF PRECOLATION RATES*

Location	Soil Tested	Depth (ft)	Percolation Rate (min/in)	Percolation Rate (gpd/sf)
P-1	Sandy Silt (ML)	4.5	2.0	448.8
P-2	Silty SAND (SM)	5.1	1.5	598.4
P-3	Silty SAND (SM)	4.8	0.9	997.4
P-4	Poorly Graded Sand with SILT (SP-SM)	5.25	0.8	1122.1

*See Appendix B for the Infiltration Feasibility Report produced by Technicon Engineering Services, Inc., dated July 6, 2016.

Due to the above percolation rates, imported sand was installed 30” below the gravel layer in order to obtain an appropriate application rate of 1.19 gpd/sf. This is consistent with Mariposa County application rate requirement of 1.2 gpd/sf.

Recycled water projects are not included in the Glacier Point Wastewater facilities, nor would they be practical due to the seasonal nature of the use and that the landscape is natural and not subject to irrigation.

Section 3: Groundwater Quality

Underlying groundwater use in the area is for domestic purposes. Industrial or agricultural uses are not permitted in the National Park and it is unlikely that these uses will ever be realized. Groundwater in this area is unconfined and levels vary widely depending on the season.

In accordance with the Infiltration Feasibility Report, high ground water is expected seasonally (late Spring/early Summer). Groundwater measurements will be done weekly at the beginning of the season and until groundwater levels are below the required level. Mariposa County requires ground water to be 5’ below the bottom of the leachfield facility, which has a total depth of 5’. Therefore, maximum allowable seasonal high groundwater is 10 feet below grade. Twelve monitoring wells are installed in and around the leachfield and NPS maintenance personnel will utilize data logging pressure transducers to monitor groundwater levels. See Figure 5 for the Leachfield Distribution System Profile.

Section 4: California Environmental Quality Act

Environmental impacts were conducted as part of the National Environmental Protection Act (NEPA), since the National Park is federally operated. See Appendix C, NEPA Compliance, for the environmental documents prepared for this project.

Section 5: Additional Technical Reports

Not applicable

END OF DOCUMENT

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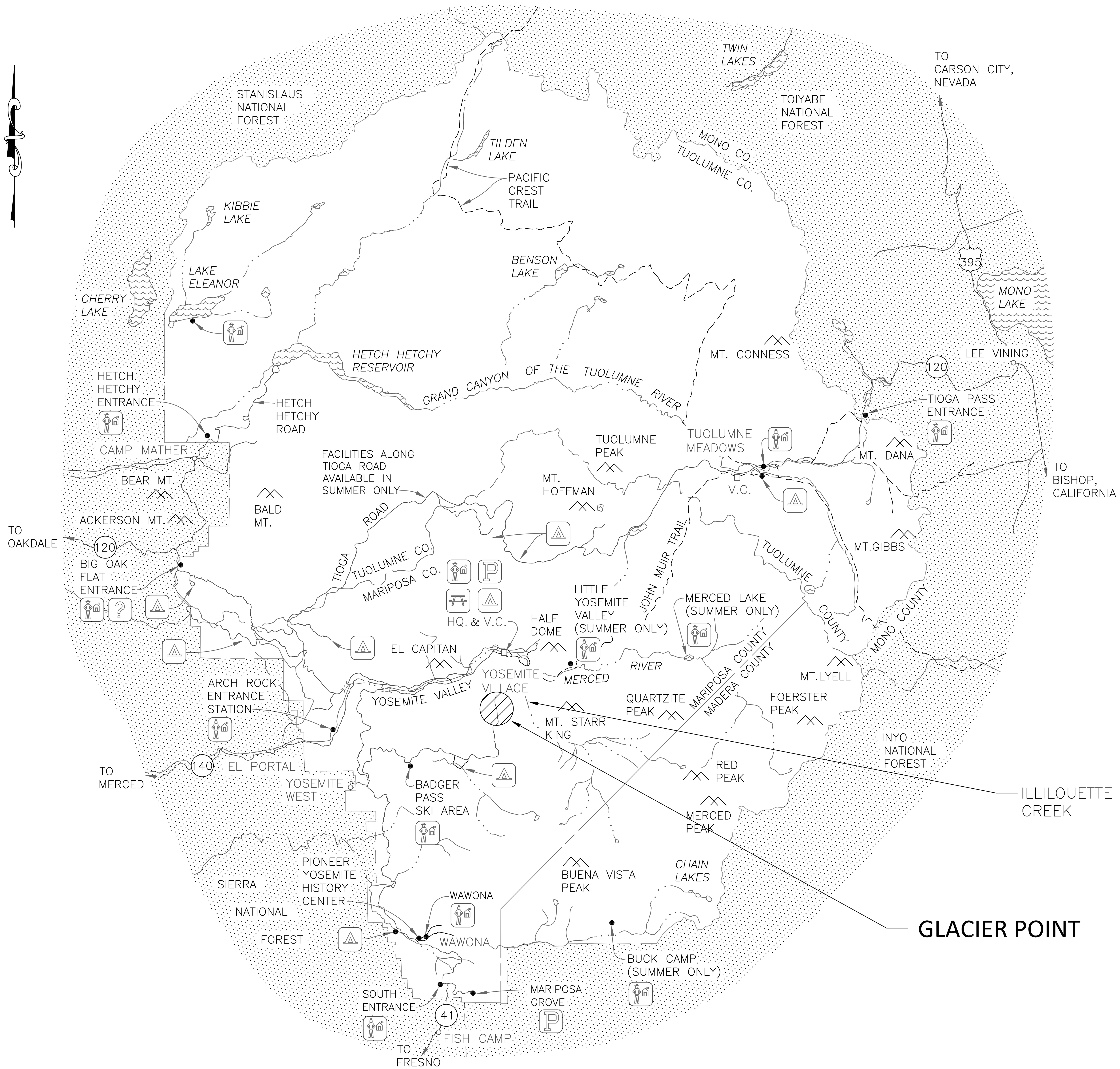
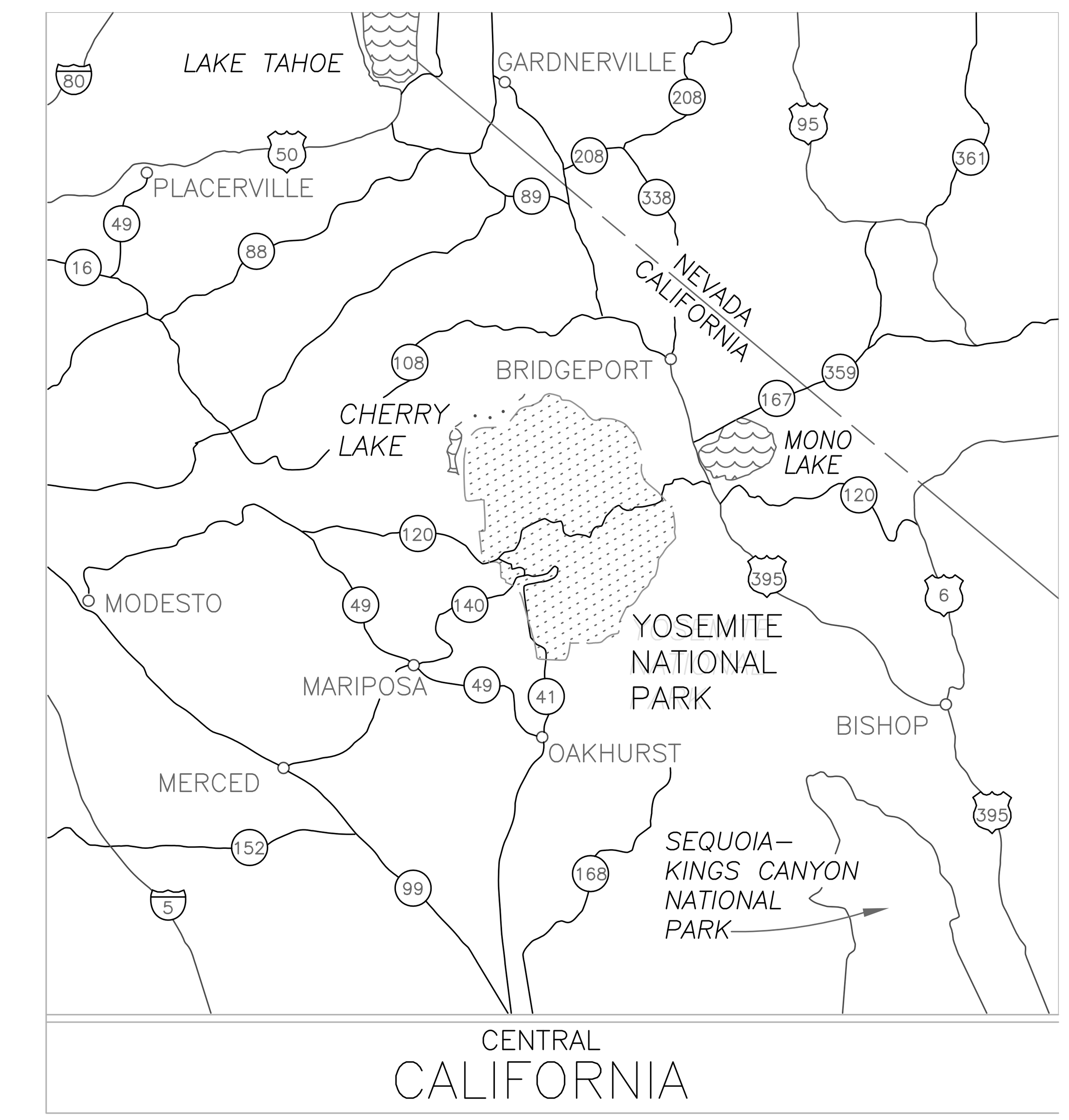
FIGURE 1 - LOCATION & VICINITY MAP



YOSEMITE NATIONAL PARK VICINITY MAP



LOCATION MAP

LEGEND

- PARK BOUNDARY
- STATE LINE
- COUNTY LINE
- PAVED ROAD
- TRAIL
- RIVER OR CREEK
- V.C. VISITOR CENTER
- HQ. PARK HEADQUARTERS
- RANGER STATION
- ? INFORMATION CENTER
- P PARKING
- PICNIC AREA
- CAMPGROUND

UNDERGROUND UTILITY LOCATIONS SHOWN ARE BASED UPON VARIOUS RECORD UTILITY MAPS IN CONJUNCTION WITH VISIBLE EVIDENCE GATHERED DURING THE TIME THE FIELD SURVEY WAS PERFORMED. THEREFORE, ALL UNDERGROUND UTILITY LOCATIONS SHOULD BE FIELD VERIFIED PRIOR TO CONSTRUCTION.

THIS DRAWING ONLY REFLECTS AND VERIFIES THE FIELD SURVEY DATA OF THOSE FEATURES AND CONDITIONS PRESENT AS OF: Date

REVISIONS	BY	DATE	HORIZONTAL DATUM: NAD 1927, CALIFORNIA STATE SYSTEM OF PLANE COORDINATES BASED ON MONUMENTS ZONE 3	UNITED STATES DEPARTMENT OF THE INTERIOR	A–E FIRM	FIELD WORK: PLP	TITLE OF SHEET	PKG. NO.	SHEET	DRAWING NO.
			VERTICAL DATUM: NGVD 1929 BASED ON MONUMENT — CONTOUR INTERVAL = 5'	NATIONAL PARK SERVICE YOSEMITE NATIONAL PARK	NAME: CITY, STATE: CONTRACT NUMBER:	DRAWN: BWF CHECKED: GC	GLACIER POINT WASTEWATER EFFLUENT SYSTEM YOSEMITE NATIONAL PARK		OF	104

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NAME:
DESCRIPTION:

N.
E.
ELEV.

FIGURE 2 - SITE MAP
GLACIER POINT

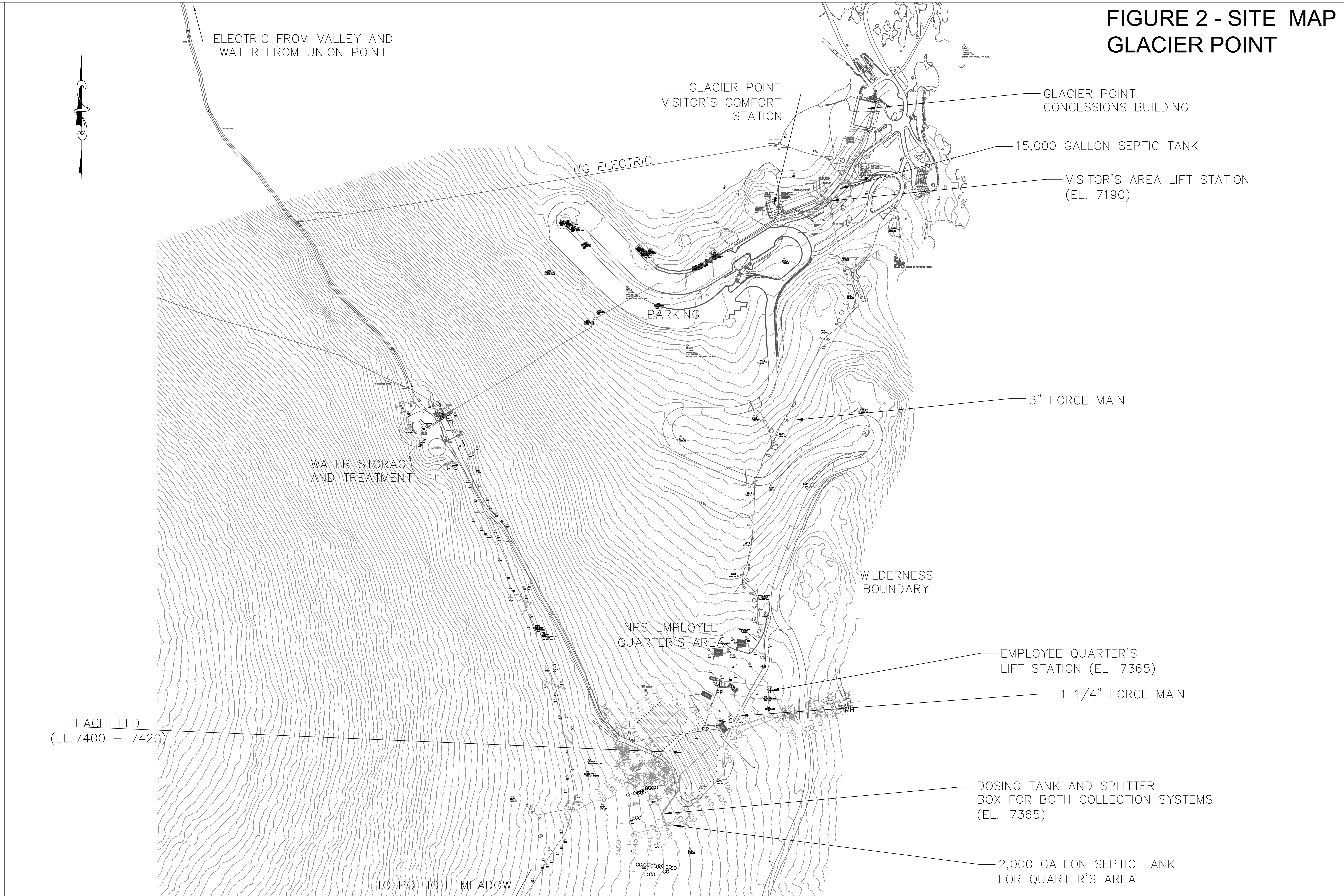

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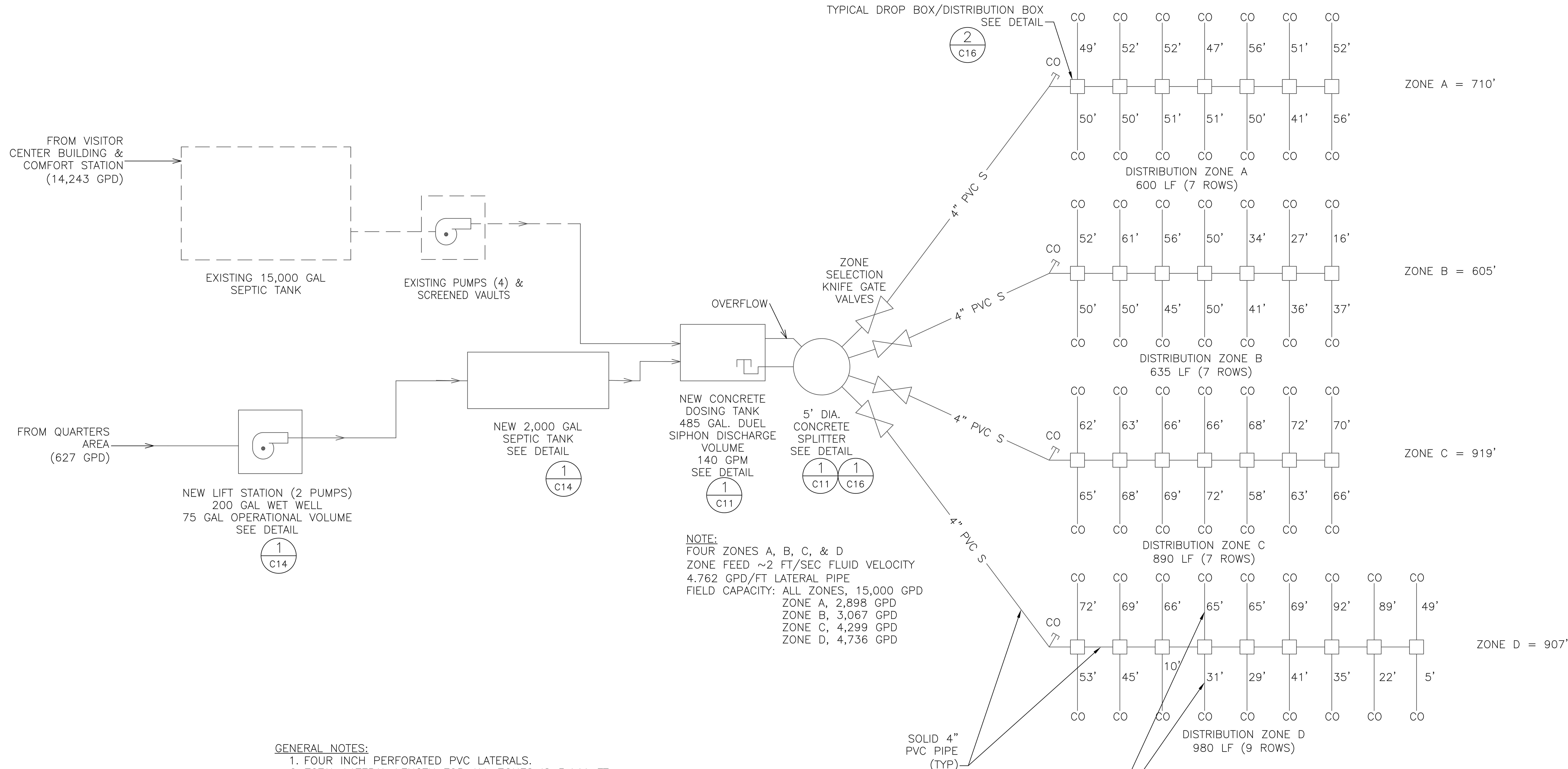
FIGURE 3 - SYSTEM FLOW DIAGRAM



NOTE:
FOUR ZONES A, B, C, & D
ZONE FEED ~2 FT/SEC FLUID VELOCITY
4.762 GPD/FT LATERAL PIPE
FIELD CAPACITY: ALL ZONES, 15,000 GPD
ZONE A, 2,898 GPD
ZONE B, 3,067 GPD
ZONE C, 4,299 GPD
ZONE D, 4,736 GPD

GENERAL NOTES:
1. FOUR INCH PERFORATED PVC LATERALS.
2. TOTAL LATERAL LENGTH FOR ALL ZONES IS 3,141 FT
3. PIPE INTERIOR VOLUME OF 2,026 GAL TOTAL ALL ZONES, CLEANOUTS AT THE END OF EACH LATERAL PIPE, (29) 4-HOLE D-BOXES TUF-TITE #4HD2 PER ZONE
4. LATERAL INVERT TWO INCHES BELOW OUTLET, OUTLET INVERT ONE INCH BELOW INLET

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FIGURE 4 - DESIGN CRITERIA

VISITOR'S CENTER SEPTIC TANK	
HISTORICAL FLOW, GPD	14,243
VOLUME REQUIRED = (FLOW)(0.75)+1,125; PER CODE, GAL.	11,807
EXISTING SEPTIC TANK, GAL.	15,000
EXISTING FOUR PUMPS AT VISITOR'S CENTER (ORENCO MODEL PF101012)	
PIPE DIAMETER, IN. (NOMINAL)	3.00
PIPE DIAMETER, IN. (ID) [HDPE SDR 9]	2.68
EXISTING FLOW FOR TWO PUMPS, GPM	20
EXISTING VELOCITY, (TWO PUMPS) FPS	1.03
EXISTING VELOCITY, (FOUR PUMPS) FPS	2.06
EXISTING TDH REQUIRED, FT. WC	218
EXISTING TDH AVAILABLE, FT. WC	255
EXISTING POWER FOR EACH PUMP, HP	1.0
BID OPTION A	
QUARTERS AREA SEPTIC TANK	
PEAK FLOW, GPD	627
VOL = FLOW X 0.75 + 1,125; PLUMBING CODE APP H 2.0	
VOLUME, GAL.	1,595
TANK DESIGN VOLUME GAL.	2,000
QUARTERS AREA LIFT STATION (ZOELLER X840 SERIES LIFT STATION)	
LIFT STATION PUMP, GPM	13.3
PIPE DIAMETER, IN. (NOMINAL)	1.66
PIPE DIAMETER, IN. (ID) [HDPE SDR 9]	1.27
FLUID VELOCITY, FPS	3.48
PIPE LENGTH, FT.	450
HAZEN WILLIAMS "C" FACTOR - HDPE	140
ELEVATION, INITIAL, FT.	7363
ELEVATION, FINAL, FT.	7428
ELEVATION CHANGE, FT.	65.0
TDH, FT.	84.5

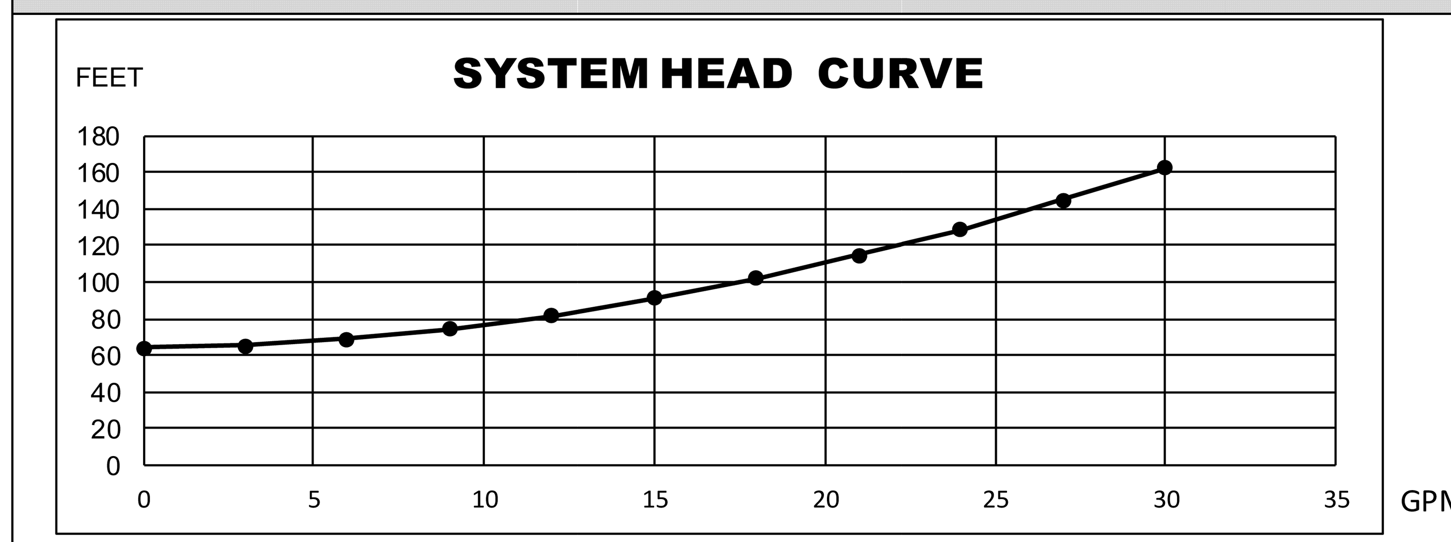
QUARTERS AREA CABINS - BID OPTION A

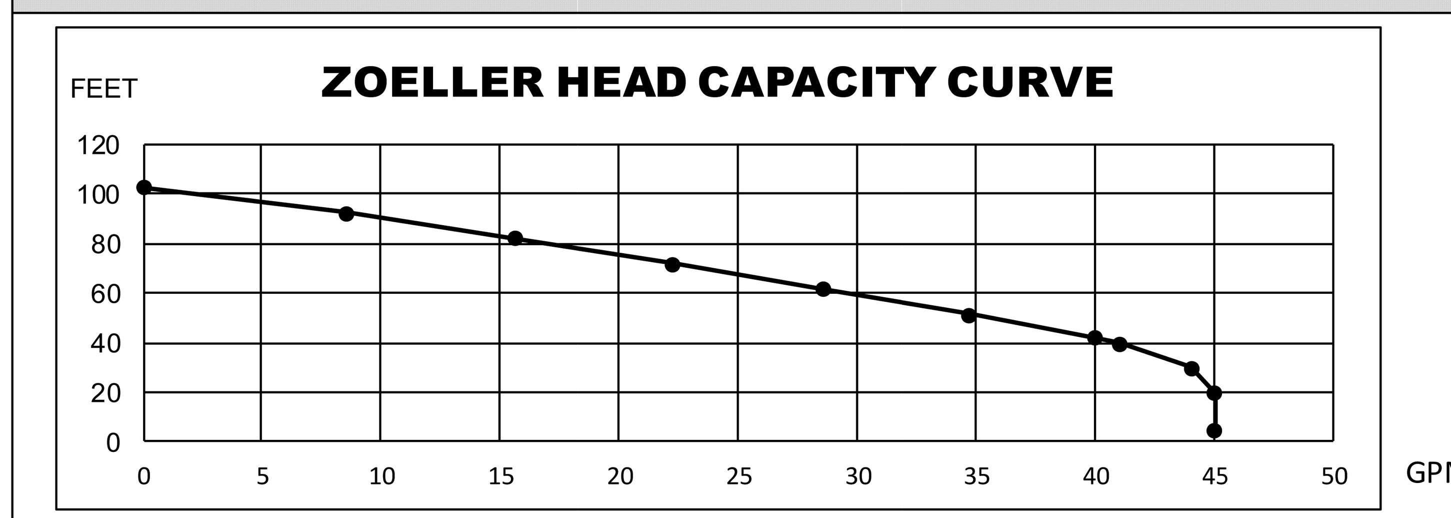
PUMPS

Item	Elevation (ft)	Pump	Hydraulic Grade (Suction) (ft)	Hydraulic Grade (Discharge) (ft)	Flow (Total) (gpm)	Pump Head (ft)	Pressure Discharge (psi)
Pump	7,363.00	ZOELLER X840	7,362.98	7,448.67	13.3	85.7	37.1

PIPE

Item	Length (ft)	Diameter (in)	Material	Hazen Williams C	Flow (gpm)	Velocity (ft/s)	Headloss (ft)
Pipe 1	1	2	SST	120	13.3	1.36	0.02
Pipe 2	10	1.25	SST	120	13.3	3.48	1.16
Pipe 3	450	1.25	HDPE	140	13.3	3.48	20.51





DRAINFIELD - SIPHON DOSING TANK	
60% TO 75% OF PIPE INTERIOR TO BE DOSED AT ONE TIME; PLUMBING CODE, APP H 6.8	
ZONES USED AT ONE TIME (VARIES)	1 TO 4
MAXIMUM ALLOWABLE LATERAL LENGTH, FT.	100
LATERAL LENGTH TOTAL, FT.	3,105
DIAMETER, IN.	4
LATERAL VOL- SOLID PIPE + PERFORATED PIPE (ZONE A), GAL.	147+391
LATERAL VOL- SOLID PIPE + PERFORATED PIPE (ZONE B), GAL.	147+413
LATERAL VOL- SOLID PIPE + PERFORATED PIPE (ZONE C), GAL.	68+580
LATERAL VOL- SOLID PIPE + PERFORATED PIPE (ZONE D), GAL.	68+638
LATERAL VOL- SOLID PIPE + PERFORATED PIPE (ALL ZONES), GAL.	430+2,022
ZONE VOLUME, GAL.	638
ALL 4 ZONES PIPE CAPACITY VOLUME, GAL.	2,452
DOSING SIZE COEFFICIENT FROM CODE	0.73
REQUIRED DOSING CAPACITY, GAL.	450
DRAINFIELD - SIPHON DOSING TANK	
DESIGN DOSING CAPACITY, GAL.	485
DESIGN DOSE FLOWRATE, GPM	140
DESIGN TANK HEIGHT; BOTTOM OF TANK TO INVERT, FT.	5.00
DESIGN WIDTH, FT.	4.92
DESIGN LENGTH, FT.	15.92
DRAW DOWN, IN.	12.0
DRAINFIELD TANKAGE - MAX DOSING FREQUENCY	
VOLUME OF WASTEWATER, GPD	15,000
DAILEY DOSES PER SELECTED DOSE VOLUME, 1/DAY	31.0
TIME BETWEEN DOSES UNDER MAXIMUM FLOW CONDITION, MIN.	16
SPLITTER PIPE EFFLUENT VELOCITIES	
ZONE A, FPS / GPM	6.62/260
ZONE B, FPS / GPM	6.62/260
ZONE C, FPS / GPM	12.2/475
ZONE D, FPS / GPM	12.2/475
GLACIER POINT NEW DRAINFIELD	
VISITOR'S CENTER SANITARY SEWER MAXIMUM FLOW, GPD	14,243
QUARTERS-AREA SANITARY SEWER MAXIMUM FLOW, GPD	627
TOTAL ESTIMATED SANITARY SEWER MAXIMUM FLOW, GPD	14,870
PERMITTED DRAINFIELD DESIGN CAPACITY, GPD	15,000
AVAILABLE REMAINING CAPACITY, GPD	130
NUMBER OF ZONES (A, B, C AND D)	4
TOTAL LATERAL LENGTH FOR ALL ZONES, FT.	3,105
DRAIN SAND FILL DEPTH UNDER PIPE, FT.	2.5
DISALLOWED SIDE HEIGHT, FT	0.5
DESIGN SIDE HEIGHT, FT	2
EFFECTIVE AREA OF APPLICATION PER FOOT OF TRENCH LENGTH, NOT INCLUDING BOTTOM (CODE REQUIRES MIN 4 SF.)	4
APPLICATION AREA FOR ALL ZONES, SF.	12,420
APPLICATION RATE, GPD/SQFT	1.19
GLACIER POINT DRAINFIELD STOCKPILING VOLUME OF DRAIN SAND	
TOTAL LATERAL FOOTAGE FOR ALL ZONES, FT.	3,105
TRENCH WIDTH, FT.	2.0
DRAIN ROCK THICKNESS, FT	1.0
ENGINEERED SAND THICKNESS FOR EFFLUENT TREATMENT, FT	2.50
TOTAL VOLUME OF DRAIN SAND, CY	575
GLACIER POINT DRAINFIELD STOCKPILE VOLUME OF FOREST DUFF & TOPSOIL	
DRAINFIELD AREA, SF.	30,949
STOCKPILE FOREST DUFF THICKNESS, FT	0.17
STOCKPILE FOREST DUFF VOLUME, CY	195
STOCKPILE TOPSOIL DEPTH, FT	0.50
STOCKPILE TOPSOIL VOLUME, CY	573

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CONTOUR INTERVAL = 5'

UNITED STATES
DEPARTMENT OF THE INTERIOR
NATIONAL PARK SERVICE
YOSEMITE NATIONAL PARK

A–E FIRM
NAME:
CITY, STATE:
CONTRACT NUMBER:

FIELD WORK: PLP
DRAWN: BWF
CHECKED: GC

TITLE OF SHEET
GLACIER POINT
WASTEWATER EFFLUENT SYSTEM
YOSEMITE NATIONAL PARK

PKG. NO.
SHEET
OF
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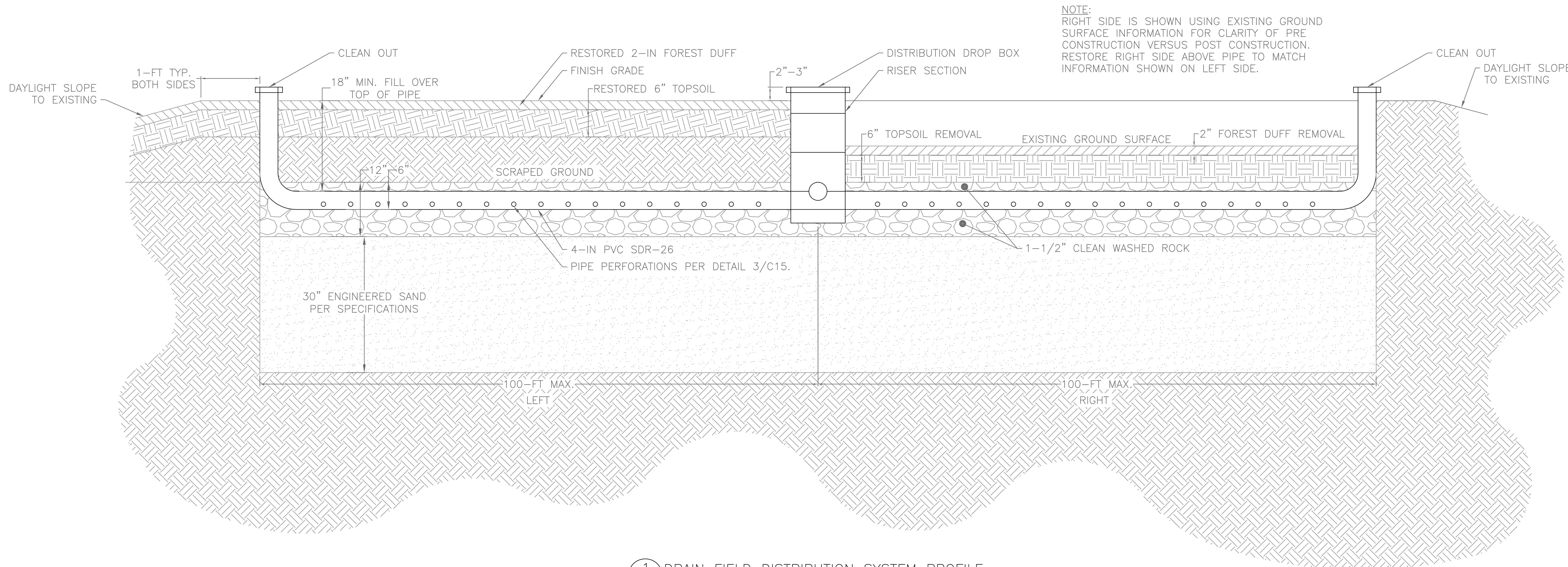
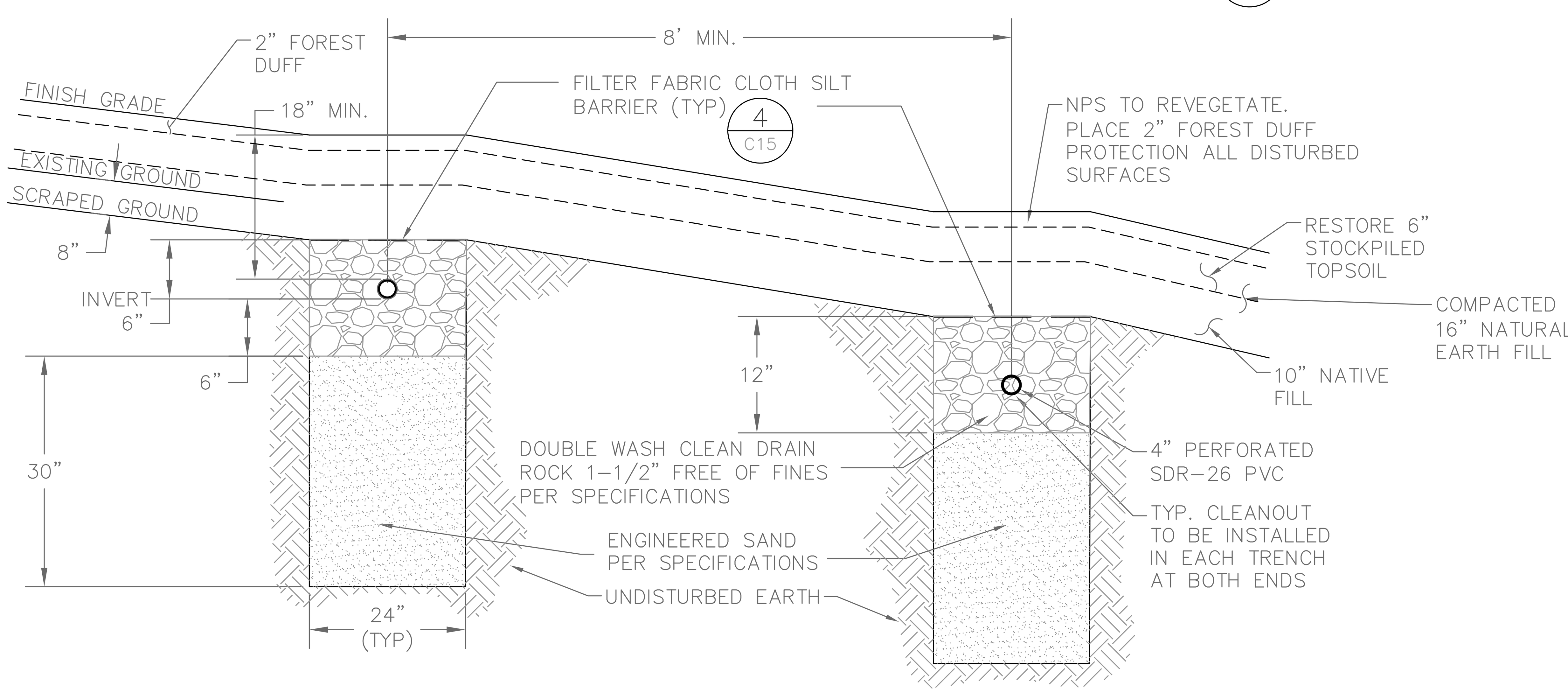
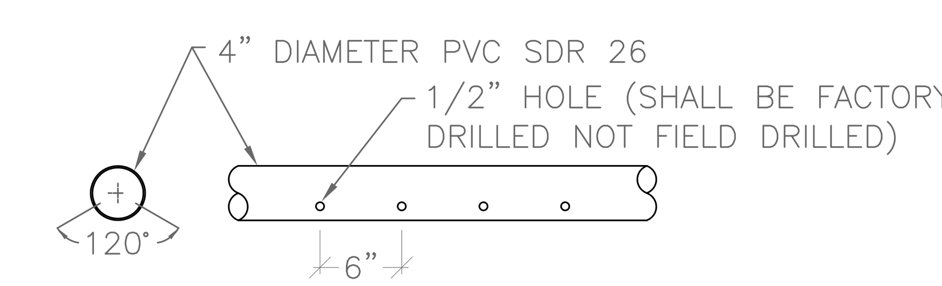
FIGURE 5 - LEACHFIELD DISTRIBUTION SYSTEM PROFILE



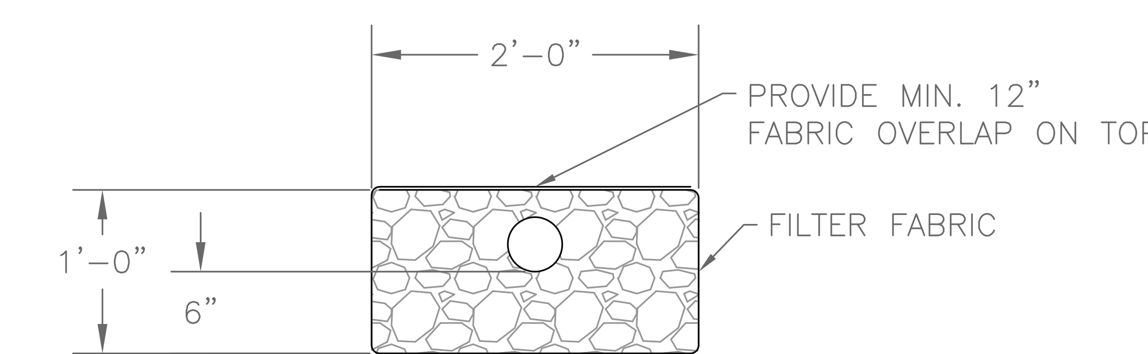
1 C15 DRAIN FIELD DISTRIBUTION SYSTEM PROFILE
NTS



2 C15 TYPICAL DRAIN FIELD SECTION
NTS



3 C15 PERFORATED PIPE
NTS



4 C15 FILTER FABRIC
NTS

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ATTACHMENT A

INTRODUCTION

This application package constitutes a Report of Waste Discharge (ROWD) pursuant to California Water Code Section 13260. Section 13260 states that persons discharging or proposing to discharge waste that could affect the quality of the waters of the State, other than into a community sewer system, shall file a ROWD containing information which may be required by the appropriate Regional Water Quality Control Board (RWQCB).

This package is to be used to start the application process for all waste discharge requirements (WDRs) and National Pollutant Discharge Elimination System (NPDES) permits* issued by a RWQCB except:

a) Those landfill facilities that must use a joint Solid Waste Facility Permit Application Form, California Integrated Waste Management Board Form E-1-77; and
b) General WDRs or general NPDES permits that use a Notice of Intent to comply or specify the use of an alternative application form designed for that permit.

This application package contains:

1. Application/General Information Form for WDRs and NPDES Permits [Form 200 (10/97)].
2. Application/General Information Instructions.

Instructions

Instructions are provided to assist you with completion of the application. If you are unable to find the answers to your questions or need assistance with the completion of the application package, please contact your RWQCB representative. *The RWQCBs strongly recommend that you make initial telephone or personal contact with RWQCB regulatory staff to discuss a proposed new discharge before submitting your application.* The RWQCB representative will be able to answer procedural and annual fee related questions that you may have. (See map and telephone numbers inside of application cover.)

All dischargers regulated under WDRs and NPDES permits must pay an annual fee, except dairies, which pay a filing fee only. The RWQCB will notify you of your annual fee based on an evaluation of your proposed discharge. Please do NOT submit a check for your first annual fee or filing fee until requested to do so by a RWQCB representative. Dischargers applying for reissuance (renewal) of an existing NPDES permit or update of an existing WDR will be billed through the annual fee billing system and are therefore requested NOT to submit a check with their application. Checks should be made payable to the State Water Resources Control Board.

Additional Information Requirements

A RWQCB representative will notify you within 30 days of receipt of the application form and any supplemental documents whether your application is complete. If your application is incomplete, the RWQCB representative will send you a detailed list of discharge specific information necessary to complete the application process. The completion date of your application is normally the date when all required information, including the correct fee, is received by the RWQCB.

*** NPDES PERMITS:** If you are applying for a permit to discharge to surface water, you will need an NPDES permit which is issued under both State and Federal law and may be required to complete one or more of the following Federal NPDES permit application forms: Short Form A, Standard Form A, Forms 1, 2B, 2C, 2D, 2E, and 2F. These forms may be obtained at a RWQCB office or can be ordered from the National Center for Environmental Publications and Information at (513) 891-6561.

CALIFORNIA ENVIRONMENTAL PROTECTION AGENCY

State of California
Regional Water Quality Control Board

APPLICATION/REPORT OF WASTE DISCHARGE GENERAL INFORMATION FORM FOR WASTE DISCHARGE REQUIREMENTS OR NPDES PERMIT

INSTRUCTIONS

FOR COMPLETING THE APPLICATION/REPORT OF WASTE DISCHARGE GENERAL INFORMATION FORM FOR: WASTE DISCHARGE REQUIREMENTS/NPDES PERMIT

If you have any questions on the completion of any part of the application, please contact your RWQCB representative. A map of RWQCB locations, addresses, and telephone numbers is located on the reverse side of the application cover.

I. FACILITY INFORMATION

You must provide the factual information listed below for ALL owners, operators, and locations and, where appropriate, for ALL general partners and lease holders.

A. FACILITY:
Legal name, physical address including the county, person to contact, and phone number at the facility.
(**NO P.O. Box numbers!** If no address exists, use street and nearest cross street.)

B. FACILITY OWNER:
Legal owner, address, person to contact, and phone number. Also include the owner's Federal Tax Identification Number.

OWNER TYPE:
Check the appropriate Owner Type. The legal owner will be named in the WDRs/NPDES permit.

C. FACILITY OPERATOR (The agency or business, not the person):
If applicable, the name, address, person to contact, and telephone number for the facility operator. Check the appropriate Operator Type. If identical to B. above, enter "same as owner".

D. OWNER OF THE LAND:
Legal owner of the land(s) where the facility is located, address, person to contact, and phone number. Check the appropriate Owner Type. If identical to B. above, enter "same as owner".

E. ADDRESS WHERE LEGAL NOTICE MAY BE SERVED:
Address where legal notice may be served, person to contact, and phone number. If identical to B. above, enter "same as owner".

F. BILLING ADDRESS
Address where annual fee invoices should be sent, person to contact, and phone number. If identical to B. above, enter "same as owner".

CALIFORNIA ENVIRONMENTAL PROTECTION AGENCY

State of California
Regional Water Quality Control Board

APPLICATION/REPORT OF WASTE DISCHARGE GENERAL INFORMATION FORM FOR WASTE DISCHARGE REQUIREMENTS OR NPDES PERMIT

II. TYPE OF DISCHARGE

Check the appropriate box to describe whether the waste will be discharged to: A. Land, or B. Surface Water.

Check the appropriate box(es) which best describe the activities at your facility.

Hazardous Waste - **If you check the Hazardous Waste box,** STOP **and contact a representative of the RWQCB for further instructions.**

Landfills - **A separate form, APPLICATION FOR SOLID WASTE FACILITY PERMIT/WASTE DISCHARGE REQUIREMENTS, California Integrated Waste Management Board Form E-1-77, may be required. Contact a RWQCB representative to help determine the appropriate form for your discharge.**

III. LOCATION OF THE FACILITY

1. Enter the Assessor's Parcel Number(s) (APN), which is located on the property tax bill. The number can also be obtained from the County Assessor's Office. Indicate the APN for both the facility and the discharge point.

2. Enter the Latitude of the entrance to the proposed/existing facility and of the discharge point. Latitude and longitude information can be obtained from a U.S. Geological Survey quadrangle topographic map. Other maps may also contain this information.

3. Enter the Longitude of the entrance to the proposed/existing facility and of the discharge point.

IV. REASON FOR FILING

NEW DISCHARGE OR FACILITY:
A discharge or facility that is proposed but does not now exist, or that does not yet have WDRs or an NPDES permit.

CHANGE IN DESIGN OR OPERATION:
A material change in design or operation from existing discharge requirements. Final determination of whether the reported change is material will be made by the RWQCB.

CHANGE IN QUANTITY/TYPE OF DISCHARGE:
A material change in characteristics of the waste from existing discharge requirements. Final determination of whether the reported change would have a significant effect will be made by the RWQCB.

CHANGE IN OWNERSHIP/OPERATOR:
Change of legal owner of the facility. Complete Parts I, III, and IV only and contact the RWQCB to determine if additional information is required.

WASTE DISCHARGE REQUIREMENTS UPDATE OR NPDES PERMIT REISSUANCE:
WDRs must be updated periodically to reflect changing technology standards and conditions. A new application is required to reissue an NPDES permit which has expired.

OTHER:
If there is a reason other than the ones listed, please describe the reason on the space provided. (If more space is needed, attach a separate sheet.)

CALIFORNIA ENVIRONMENTAL PROTECTION AGENCY

State of California
Regional Water Quality Control Board

APPLICATION/REPORT OF WASTE DISCHARGE GENERAL INFORMATION FORM FOR WASTE DISCHARGE REQUIREMENTS OR NPDES PERMIT

V. CALIFORNIA ENVIRONMENTAL QUALITY ACT (CEQA)

It should be emphasized that communication with the appropriate RWQCB staff is vital before starting the CEQA documentation, and is recommended before completing this application. There are Basin Plan issues which may complicate the CEQA effort, and RWQCB staff may be able to help in providing the needed information to complete the CEQA documentation.

Name the Lead Agency responsible for completion of CEQA requirements for the project, i.e., completion and certification of CEQA documentation.

Check YES or NO. Has a public agency determined that the proposed project is exempt from CEQA?
If the answer is YES, state the basis for the exemption and the name of the agency supplying the exemption on the space provided. (Remember that, if extra space is needed, use an extra sheet of paper, but be sure to indicate the attached sheet under Section VII. Other.)

Check YES or NO. Has the "Notice of Determination" been filed under CEQA? If YES, give the date the notice was filed and enclose a copy of the Notice of Determination and the Initial Study, Environmental Impact Report, or Negative Declaration. If NO, check the box of the expected type of CEQA document for this project, and include the expected date of completion using the timelines given under CEQA. The date of completion should be taken as the date that the Notice of Determination will be submitted. (If not known, write "Unknown")

VI. OTHER REQUIRED INFORMATION

To be approved, your application MUST include a COMPLETE characterization of the discharge. If the characterization is found to be incomplete, RWQCB staff will contact you and request that additional specific information be submitted.

This application MUST be accompanied by a site map. A USGS 7.5' Quadrangle map or a street map, if more appropriate, is sufficient for most applications.

VII. OTHER

If any of the answers on your application form need further explanation, attach a separate sheet. Please list any attachments with the titles and dates on the space provided.

VIII. CERTIFICATION

Certification by the owner of the facility or the operator of the facility, if the operator is different from the owner, is required. The appropriate person must sign the application form.
Acceptable signatures are:

1. **for a corporation,** a principal executive officer of at least the level of senior vice-president;
2. **for a partnership or individual (sole proprietorship),** a general partner or the proprietor;
3. **for a governmental or public agency,** either a principal executive officer or ranking elected/appointed official.

DISCHARGE SPECIFIC INFORMATION

In most cases, a request to supply additional discharge specific information will be sent to you by a representative of the RWQCB. If the RWQCB determines that additional discharge specific information is not needed to process your application, you will be so notified.

CALIFORNIA ENVIRONMENTAL PROTECTION AGENCY

State of California
Regional Water Quality Control Board

APPLICATION/REPORT OF WASTE DISCHARGE GENERAL INFORMATION FORM FOR WASTE DISCHARGE REQUIREMENTS OR NPDES PERMIT

I. FACILITY INFORMATION

A. Facility:

Name: Glacier Point Wastewater Facility			
Address: 5205 Glacier Point Road			
City: Yosemite National Park	County: Mariposa	State: CA	Zip Code: 95389
Contact Person: Rick Hall		Telephone Number: 209-379-1077	

B. Facility Owner:

Name: National Parks Service Department of the Interior			Owner Type (Check One): 1. ☐ Individual 2. ☐ Corporation
Address: PO Box 577			3. ☑ Governmental Agency 4. ☐ Partnership
City: Yosemite NP	State: CA	Zip Code: 95389	5. ☐ Other:
Contact Person: Cicely Muldoon		Telephone Number: 209-372-0201	Federal Tax ID: Not Applicable

C. Facility Operator (The agency or business, not the person):

Name: NPS DOI			Operator Type (Check One): 1. ☐ Individual 2. ☐ Corporation
Address: PO Box 700			3. ☑ Governmental Agency 4. ☐ Partnership
City: El Portal	State: CA	Zip Code: 95318	5. ☐ Other:
Contact Person: Rick Hall		Telephone Number: 209-379-1077	

D. Owner of the Land:

Name: NPS DOI			Owner Type (Check One): 1. ☐ Individual 2. ☐ Corporation
Address: PO Box 577			3. ☑ Governmental Agency 4. ☐ Partnership
City: Yosemite NP	State: CA	Zip Code: 95389	5. ☐ Other:
Contact Person: Cicely Muldoon		Telephone Number: 209-372-0201	

E. Address Where Legal Notice May Be Served:

Address: PO Box 577		
City: Yosemite NP	State: CA	Zip Code: 95389
Contact Person: Cicely Muldoon		Telephone Number: 209-372-0201

F. Billing Address:

Address: PO Box 700		
City: El Portal	State: CA	Zip Code: 95318
Contact Person: Rick Hall		Telephone Number: 209-379-1077

CALIFORNIA ENVIRONMENTAL PROTECTION AGENCY

State of California
Regional Water Quality Control Board

APPLICATION/REPORT OF WASTE DISCHARGE
GENERAL INFORMATION FORM FOR
WASTE DISCHARGE REQUIREMENTS OR NPDES PERMIT

II. TYPE OF DISCHARGE

Check Type of Discharge(s) Described in this Application (A or B):

☑ **A. WASTE DISCHARGE TO LAND** ☐ **B. WASTE DISCHARGE TO SURFACE WATER**

Check all that apply:

- ☑ Domestic/Municipal Wastewater Treatment and Disposal
- ☐ Cooling Water
- ☐ Mining
- ☐ Waste Pile
- ☐ Wastewater Reclamation
- ☐ Animal Waste Solids
- ☐ Land Treatment Unit
- ☐ Dredge Material Disposal
- ☐ Surface Impoundment
- ☐ Industrial Process Wastewater
- ☐ Animal or Aquacultural Wastewater
- ☐ Biosolids/Residual
- ☐ Hazardous Waste (see instructions)
- ☐ Landfill (see instructions)
- ☐ Storm Water
- ☑ Other, please describe: Employee Cabins, Visitor Restrooms

III. LOCATION OF THE FACILITY

Describe the physical location of the facility.

1. Assessor's Parcel Number(s)	**2. Latitude**	**3. Longitude**
Facility:	**Facility:** 27.724102	**Facility:** W 119.574561
Discharge Point:	**Discharge Point:** SAME	**Discharge Point:** SAME

IV. REASON FOR FILING

- ☑ New Discharge or Facility
- ☑ Change in Design or Operation
- ☑ Change in Quantity/Type of Discharge
- ☐ Changes in Ownership/Operator (see instructions)
- ☐ Waste Discharge Requirements Update or NPDES Permit Reissuance
- ☐ Other: ____________________

V. CALIFORNIA ENVIRONMENTAL QUALITY ACT (CEQA)

Name of Lead Agency: National Parks Service, Department of the Interior

Has a public agency determined that the proposed project is exempt from CEQA? ☐ **Yes** ☑ **No**

If Yes, state the basis for the exemption and the name of the agency supplying the exemption on the line below.

Basis for Exemption/Agency: ____________________

Has a "Notice of Determination" been filed under CEQA? ☐ **Yes** ☑ **No**

If Yes, enclose a copy of the CEQA document, Environmental Impact Report, or Negative Declaration. If no, identify the expected type of CEQA document and expected date of completion.

Expected CEQA Documents:

See enclosed NEPA docs. Regional Water Board is the only state agency involved; request a CEQA determination.

☐ **EIR** ☐ **Negative Declaration**

Expected CEQA Completion Date: ____________________

CALIFORNIA ENVIRONMENTAL PROTECTION AGENCY

State of California
Regional Water Quality Control Board

APPLICATION/REPORT OF WASTE DISCHARGE GENERAL INFORMATION FORM FOR WASTE DISCHARGE REQUIREMENTS OR NPDES PERMIT

VI. OTHER REQUIRED INFORMATION

Please provide a COMPLETE characterization of your discharge. A complete characterization includes, but is not limited to, design and actual flows, a list of constituents and the discharge concentration of each constituent, a list of other appropriate waste discharge characteristics, a description and schematic drawing of all treatment processes, a description of any Best Management Practices (BMPs) used, and a description of disposal methods.

Also include a site map showing the location of the facility and, if you are submitting this application for an NPDES permit, identify the surface water to which you propose to discharge. Please try to limit your maps to a scale of 1:24,000 (7.5' USGS Quadrangle) or a street map, if more appropriate.

VII. OTHER

Attach additional sheets to explain any responses which need clarification. List attachments with titles and dates below:

You will be notified by a representative of the RWQCB within 30 days of receipt of your application. The notice will state if your application is complete or if there is additional information you must submit to complete your Application/Report of Waste Discharge, pursuant to Division 7, Section 13260 of the California Water Code.

VIII. CERTIFICATION

"I certify under penalty of law that this document, including all attachments and supplemental information, were prepared under my direction and supervision in accordance with a system designed to assure that qualified personnel properly gathered and evaluated the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment."

Print Name: Cicely Muldoon　　Title: Superintendent, Yosemite NP

Signature: [signature]　　Date: MAR 24 2020

FOR OFFICE USE ONLY

Date Form 200 Received:	Letter to Discharger:	Fee Amount Received:	Check #:

California Environmental Protection Agency
Bill of Rights for Environmental
Permit Applicants

California Environmental Protection Agency (Cal/EPA) recognizes that many complex issues must be addressed when pursuing reforms of environmental permits and that significant challenges remain. We have initiated reforms and intend to continue the effort to make environmental permitting more efficient, less costly, and to ensure that those seeking permits receive timely responses from the boards and departments of the Cal/EPA. To further this goal, Cal/EPA endorses the following precepts that form the basis of a permit applicant's "Bill of Rights."

1. Permit applicants have the right to assistance in understanding regulatory and permit requirements. All Cal/EPA programs maintain an Ombudsman to work directly with applicants. Permit Assistance Centers located throughout California have permit specialists from all the State, regional, and local agencies to identify permit requirements and assist in permit processing.

2. Permit applicants have the right to know the projected fees for review of applications, how any costs will be determined and billed, and procedures for resolving any disputes over fee billings.

3. Permit applicants have the right of access to complete and clearly written guidance documents that explain the regulatory requirements. Agencies must publish a list of all information required in a permit application and of criteria used to determine whether the submitted information is adequate.

4. Permit applicants have the right of timely completeness determinations for their applications. In general, agencies notify the applicant within 30 days of any deficiencies or determine that the application is complete. California Environmental Quality Act (CEQA) and public hearing requests may require additional information.

5. Permit applicants have the right to know exactly how their applications are deficient and what further information is needed to make their applications complete. Pursuant to California Government code Section 65944, after an application is accepted as complete, an agency may not request any new or additional information that was not specified in the original application.

6. Permit applicants have the right of a timely decision on their permit application. The agencies are required to establish time limits for permit reviews.

7. Permit applicants have the right to appeal permit review time limits by statute or administratively that have been violated without good cause. For state environmental agencies, appeals are made directly to the Cal/EPA Secretary or to a specific board. For local environmental agencies, appeals are generally made to the local governing board or, under certain circumstances, to Cal/EPA. Through this appeal, applicants may obtain a set date for a decision on their permit and, in some cases, a refund of all application fees (ask boards and departments for details).

8. Permit applicants have the right to work with a single lead agency where multiple environmental approvals are needed. For multiple permits, all agency actions can be consolidated under a lead agency. For site remediation, all applicable laws can be administered through a single agency.

9. Permit applicants have the right to know who will be reviewing their application and the time required to complete the full review process.

APPENDIX A

YOSEMITE NATIONAL PARK

PMIS No. 198897A

IMPROVE THE GLACIER POINT RESTROOM
WASTEWATER EFFLUENT DISPOSAL SYSTEM

ENGINEERING DESIGN CALCULATIONS

REGISTERED PROFESSIONAL ENGINEER
DAVID BARTON BROOKE
NO. 70417
EXP. 9/30/18
CIVIL
STATE OF CALIFORNIA

David Barton Brooke

February 14, 2018

NATIONAL PARK SERVICE
PACIFIC WEST REGION

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DESCRIPTION **NO. OF PAGES**

GLACIER POINT DRAINFIELD TRENCH DESIGN CALCULATIONS

All calculations to follow are in conformance with appropriate codes and regulations as discussed in the Title I Predesign Report prepared for the Glacier Point Wastewater Improvements. Local oversight of onsite wastewater treatment system permitting, and installation inspection is administered through the Environmental Health Division of the Mariposa County Health Department, as approved by the Regional Water Quality Board. Title 13 of the county code, specifically sections 06 (sewer regulations) and 08 (sewage disposal), have the regulations and details governing sewer systems, septic tank treatment and effluent disposal.

These county regulations show that the county board can set minimum construction standards for the health officer to use in inspection and granting of permits for septic and disposal systems. These regulations also specify that the design of the systems should also be in accordance with Uniform Plumbing Codes in effect. The building department references the 2016 edition of the California Building Codes (effective January 1, 2017) as governing activity in Mariposa County. The proper section of this adopted version of codes applicable to wastewater disposal systems is the Title 24 Part 5 California Plumbing Code which is based upon the 2015 Uniform plumbing Code.

One specific criteria for drainfield design is contained in the Mariposa County Sewage Disposal Rules, Section-.080 (B) (1), saying drainfield trench absorption area is based on the infiltration of the sidewall area adjacent to the drain rock and below the distribution pipe only. This is more restrictive than what the US EPA Design Manual which indicates that the trench bottom can be included as well.

In order to create a suitable drainfield, material must be imported for use in the drainfield trench construction to achieve the minimum 5 minutes per inch (MPI) percolation rate to comply with regional and county codes. The existing soils have percolation rates at faster than 2 MPI and are thus not directly suitable to use for the current application. Assuming an actual percolation rate of the imported material of between 5 and 6 MPI to meet the county codes, the 1980 US EPA Manual "*Design Manual – Onsite Wastewater Treatment and Disposal Systems"* recommends using a less restrictive application rate of 1.2 gpd/ft^2.

Each of the drainfield trenches will be assumed to have a 24-inch width with a minimum 18-inch depth of earth cover over the lines. The design will assume 2 inches of filter media above a 4-inch diameter drain line (a total depth of 6 inches above the bottom of the drain pipe) and a minimum of 30 inches of the sand media below the drain pipe. There will be an 8-foot separation between trenches.

With this cross-sectional design, the US EPA manual indicates that both the bottom and sides of the trench below the drain pipe could be considered as infiltrative surfaces. This nominally gives a total of 7ft^2 of infiltrative surface per lineal foot of trench length. The manual also states that it is customary practice to not give credit for at least the first 6 inches of each sidewall due to clogging issues with the percolation of wastewater containing solids, thus reducing the net area to 6 ft^2 per lineal foot of trench. Due to the speed of percolation of the fine sand, conservatively it is assumed that the 6 inches of each sidewall is disallowed since some of the sidewall at the level of the drain pipe may not be used at all unless or until the bottom becomes so clogged that it ponds the water in the trench up to the level of the drain pipe.

The Mariposa Sewer Rules dictates that only side wall surfaces can be used as part of the infiltrative surface, which reduces the available area from 6 to 4 ft^2 per lineal foot of trench. Finally, the OWTS Policy further states that the maximum area to use per lineal foot of trench should be 4 ft^2. With the current design, this limit is the resultant net area and is used in further calculations. This design will then use 4 effective ft^2 per foot of trench to calculate the length of trench and disposal piping to be constructed. At the 1.2 gpd/ft^2 application rate, this equates to 4.8 gpd per linear foot of trench.

For the selected design flowrate of 15,000 GPD, dividing by the 4.8 gpd/ft trench gives a total minimum trench length of 3,125 lineal feet of trench. Using the mandated maximum 100-foot-long trench length, the length of piping needed would equate to 3,200 feet or 32 – 100-foot long trenches.

SPLITTER BOX PIPE DISCHARGE CAPACITIES

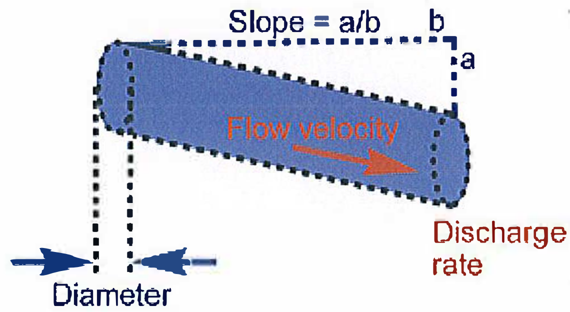
PIPE DISCARGE CAPACITY FOR ZONES A & B

MyCT
Main
Forum

Gravity-fed pipe flow

Hazen-Williams formula for a full pipe.

Search
Engineering index
Civil Engineering index



the calc

The gravitational flow form of the Hazen-Williams equation is calculated to provide water velocity and discharge rate that can be achieved through a pipe with provided proportions.

Pipe diameter:	4	inches
Roughness coefficient:	150	
Pipe length (b):	225	feet
Drop (a):	7.58	feet
Velocity:	6.62021	feet/second
Discharge rate	259.295	gal(US)/min
Pipe slope:	0.0336889	

Calculate!

Add

notes

This calc is mainly for pipes full with water at ambient temperature and under turbulent flow.
If you know the slope rather than the pipe length and drop, then enter "1" in "Length" and enter the slope in "Drop". If the cunduit is not a full circular pipe, but you know the hydraulic radius, then enter (Rh×4) in "Diameter".
Typical values of the roughness (friction loss) coefficient include: 100 (concrete, cast iron); 120 (steel); 140 (cement); 150 (copper, plastics).

Share / Save ...

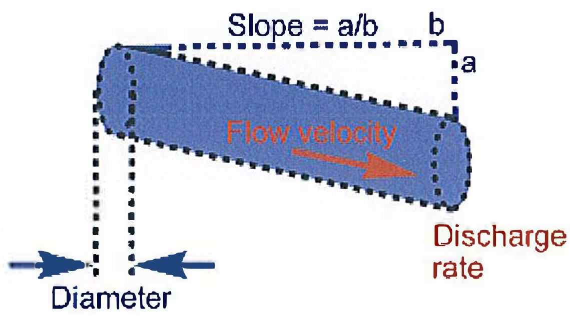
PIPE DISCHARGE CAPACITY FOR ZONES C & D

MyCT
Main
Forum

Gravity-fed pipe flow

Hazen-Williams formula for a full pipe.

Search
Engineering index
Civil Engineering index



the calc

The gravitational flow form of the Hazen-Williams equation is calculated to provide water velocity and discharge rate that can be achieved through a pipe with provided proportions.

Pipe diameter:	4	inches
Roughness coefficient:	150	
Pipe length (b):	100	feet
Drop (a):	10.38	feet
Velocity:	12.1556	feet/second
Discharge rate	476.099	gal(US)/min
Pipe slope:	0.103800	

Calculate!

Add

notes

This calc is mainly for pipes full with water at ambient temperature and under turbulent flow.
If you know the slope rather than the pipe length and drop, then enter "1" in "Length" and enter the slope in "Drop". If the cunduit is not a full circular pipe, but you know the hydraulic radius, then enter (Rh×4) in "Diameter".
Typical values of the roughness (friction loss) coefficient include: 100 (concrete, cast iron); 120 (steel); 140 (cement); 150 (copper, plastics).

Share / Save

LIFT STATION PUMP HYDRAULIC CALCULATIONS
ZOELLER X840 PUMPS

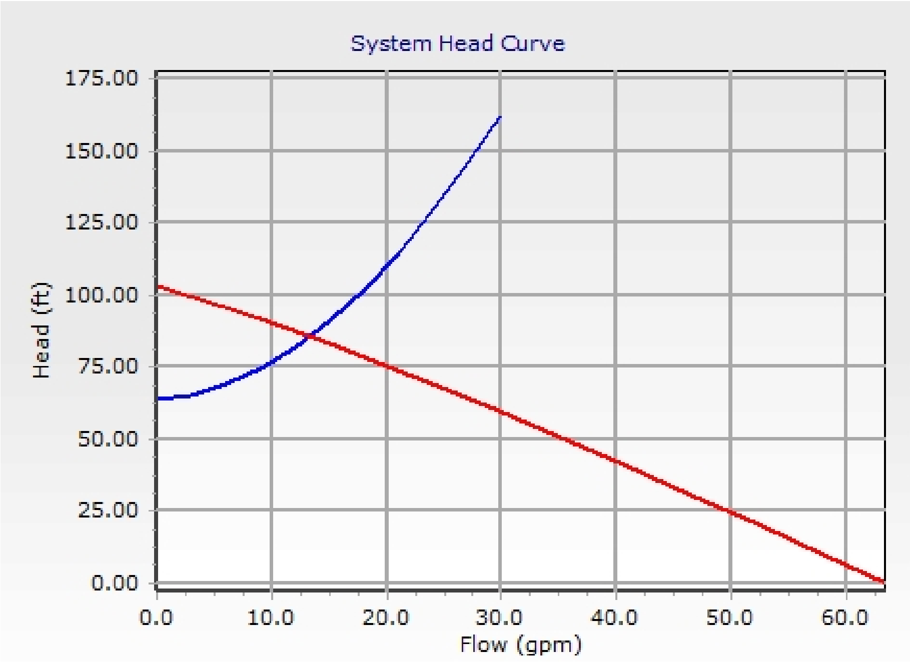
System Head Curve Detailed Report - System Head Curve - 2

Element Details

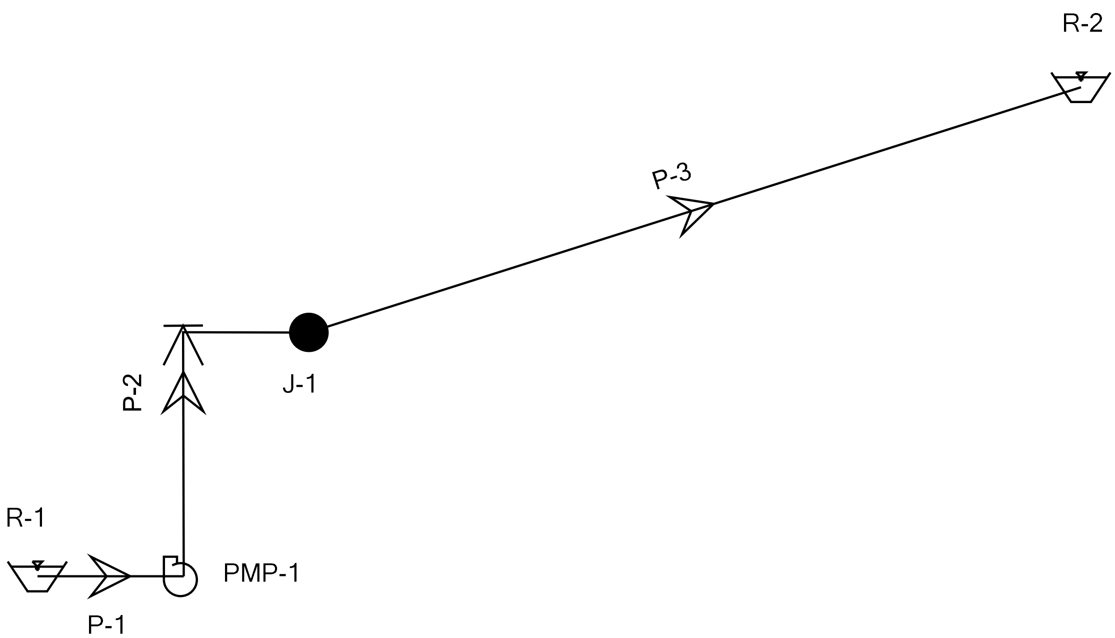
Label	System Head Curve - 2	Maximum Flow	30.0 gpm
Pump	PMP-1	Number of Intervals	10

Time (hours)
0.000

0.000 hours Flow (gpm)	0.000 hours Head (ft)	Pump Definition - Zoeller X840 Flow (gpm)	Pump Definition - Zoeller X840 Head (ft)
0.0	64.00	63.6	0.00
3.0	65.36	58.0	10.28
6.0	68.94	52.3	20.56
9.0	74.49	46.6	30.84
12.0	81.90	40.7	41.12
15.0	91.10	34.7	51.40
18.0	102.02	28.6	61.68
21.0	114.62	22.2	71.96
24.0	128.87	15.6	82.24
27.0	144.74	8.5	92.52
30.0	162.20	0.0	102.80



Scenario: Base



FlexTable: Junction Table

Label	Elevation (ft)	Zone	Demand (gpm)	Hydraulic Grade (ft)	Pressure (psi)
J-1	7,370.00	<None>	0.0	7,447.51	33.5

FlexTable: Reservoir Table

Label	Elevation (ft)	Flow (Out net) (gpm)	Hydraulic Grade (ft)
R-1	7,363.00	13.3	7,363.00
R-2	7,427.00	-13.3	7,427.00

FlexTable: Pump Table

Label	Pump Definition	Elevation (ft)	Hydraulic Grade (Suction) (ft)	Pressure (Suction) (psi)	Hydraulic Grade (Discharge) (ft)	Flow (Total) (gpm)	Pump Head (ft)	Pressure (Discharge) (psi)
PMP-1	Pump Definition - Zoeller X840	7,363.00	7,362.98	0.0	7,448.67	13.3	85.7	37.1

FlexTable: Pipe Table

Label	Length (ft)	Diameter (in)	Material	Hazen-Williams C	Minor Loss Coefficient (Local)	Flow (gpm)	Velocity (ft/s)	Headloss (ft)
P-1	1.00	2.00	Steel	120.0	0.500	13.3	1.36	0.02
P-2	10.00	1.25	Steel	120.0	3.000	13.3	3.48	1.16
P-3	450.00	1.25	HDPE	140.0	1.000	13.3	3.48	20.51

TECHNICON
ENGINEERING SERVICES, INC.
GEOTECHNICAL & ENVIRONMENTAL ENGINEERING - CONSTRUCTION TESTING & INSPECTION
1989 ♦ 2014 25 YEARS STRONG

July 6, 2016

TES No. 160577.001
Invoice No. 10763

Mr. Quin Clements
Davido Consulting Group, Inc.
1796 Main St. #105
Freeland, Washington 98249
Email: quin@dcgengr.com

Project: Glacier Point
Yosemite Valley, California

Subject: Infiltration Feasibility for Glacier Point Leach Fields

Dear Mr. Clements:

TECHNICON Engineering Services, Inc., (TECHNICON) prepared this report to summarize the installation of piezometers, infiltration testing results, and infiltration feasibility for the above referenced project located at Glacier Point in Yosemite Valley, California. The location of the site is shown on the attached Vicinity map, Figure 1.

This report includes information and test results by **TECHNICON** as well as information from a report prepared by HWA GeoSciences, Inc. (reference File HWA Project No. 2014-123 T200, dated June 14, 2016). Information from HWA GeoSciences, Inc.'s report is included with indentation herein and is cited at the end of each section of this report.

PROJECT UNDERSTANDING

Based on verbal discussions and emails with National Park Service (NPS) and HWA GeoSciences Inc., as well as a site visit, it is understood that NPS is assessing the feasibility of a proposed location for a new leach field for restroom facilities at Glacier Point. The existing restroom facilities at Glacier Point are currently pit toilets and the NPS plans to upgrade them to flushing toilets. A current leach field is located nearby to service one flushing restroom, but the leach field is not large enough to service the proposed facilities.

The purpose of the investigation was to observe and record the subsurface soil conditions, install piezometers for measuring perched groundwater conditions, and perform percolation tests to assess feasibility of creating a new leach field to service more restroom facilities.

Two (2) locations selected by the NPS, near the current leach field and at the old leach field, were tested for the feasibility of constructing a new leach field. The location near the existing leach field to the north of Glacier Point and was tested to possibly improve and expand on the existing leach field. The second area was an old spray field located east of the Glacier Point amphitheater approximately 1,000 feet down the hillside.

PROJECT BACKGROUND

Current Drainfield

Facilities served by the septic system include a comfort station and a day-use concessionaire building. The septic system was mostly built in 1997 and consists of a 15,000 gallon septic tank and wet well, pumping approximately 2,000 feet upslope to the west to a 1,000 lineal foot, 40-inch deep leachfield and a shallow, 18-inch leachfield (installed in 2000 at the same location). Due to seasonal high ground water conditions, a cutoff trench 15 feet deep had been installed in 1999 upslope from the 1997 leachfield.

The sewage treatment facilities operate during the late spring to fall months, with average wastewater flow rates ranging from 3,700 to 7,300 gallons per day with peak days of approximately 8,000 to 10,000 gallons per day (Winzler & Kelly, 2006).

Previous investigations were conducted by Melvin C. Simons Associates and Provost & Pritchard, Inc. in 1995 (National Park Service, 1995) and Winzler & Kelly Consulting Engineers in 2006. The drainfield area experiences seasonal high groundwater, typically during spring and early summer. This was the case before and after improvements in 1999 and 2000, and data from 16 monitoring wells indicate groundwater mounding occurs during peak demand times such as weekends and holidays. The Winzler & Kelly report assumed a drainfield application rate of 3 gpd/sf.

Winzler and Kelly concluded that site soils could not be used for treatment, such that treatment facilities would be necessary for on-site infiltration, such as membrane filtration, a recirculating sand or gravel filter, or a textile filter. (HWA GeoSciences, Inc., 2016)

Old Drainfield / Sprayfield

Prior wastewater facilities, from the former Glacier Point Hotel (which burned down in 1969), include a discharge pipeline to approximately 180 lineal feet of leach line located approximately 1,450 feet east. The pipeline and former leach line are inaccessible by vehicles. An Imhoff tank dating to 1932 and associated sludge drying beds midway between the septic tanks and the drainfield had been abandoned. A sprayfield irrigation line has also been abandoned. (HWA GeoSciences, Inc., 2016)

GEOLOGY

General geologic information about the Park was obtained from *The Geologic Story of Yosemite National Park* by N. King Huber (1987) and a published 1:125,000 map entitled *Geologic Map of Yosemite National Park, California* (Huber et al, 1989). According to Huber, much of the park geology consists of granitic rocks from multiple magmatic intrusions during the Cretaceous. Paleozoic metamorphic belts of rock are present to the west and east of the exposed granitic batholiths. Bedrock in the park was exposed through broad asymmetric uplift of the Sierra Mountain Range, characterized by a broad western-facing massif abruptly terminated east of the high peaks by rangefront faulting that is dropping the crust to the east. Fluvial networks developed on the western slope from long-term erosion, with multiple ages of erosional-depositional terraces documenting the progressive uplift. At least three glaciations during the Pleistocene resulted in ice caps and valley glaciers that modified the landscape. Heavily weathered bedrock was typically stripped away exposing the “fresh” rock one sees in Yosemite Valley in particular. The latest glaciation in the Sierras reached its maximum approximately 15,000 years ago and thereafter retreated. Glacier Point was glaciated during an older glaciation, but not during the latest glaciation in which ice remained below it in Yosemite Valley.

According to the 1989 geologic map, the Glacier Point study area is mapped within the Tuolumne Intrusive Suite, from magmatic intrusions that occurred during the Cretaceous. It consists specifically of Kuna Point Granodiorite, dated at approximately 88 million years.

A Geologic map of the site vicinity indicates the entire upland of which Glacier Point is the easternmost point was covered with an ice cap during Pleistocene alpine glaciations, with valley glaciers below in Yosemite Valley (Huber, 1989). (HWA GeoSciences, Inc., 2016)

SITE DESCRIPTIONS

Glacier Point Current Drainfield

The proposed drainfield area is immediately downslope of the existing drainfield and immediately upslope of the ranger cabins, reached from a dirt service road located a short distance up the road from Glacier Point at an elevation of approximately 7,400 feet. This area slopes to the northeast at approximately 10 to 20 degrees, and is forested with mature hybrid Ponderosa Pine up to approximately 48” dbh and true firs. Steeper slopes above it extend from Sentinel Dome to the west. (HWA GeoSciences, Inc.,2016)

Glacier Point Old Drainfield

This area is located downslope to the east from the Glacier Point parking lot, at an elevation just above 6,800 feet (about 400 feet lower than Glacier Point). The proposed drainfield area slopes gently east-northeast at approximately 5 to 10 degrees, and is vegetated with brush and widely spaced pines and firs. It is at the toe of steeper slopes coming down from the Glacier Point road and the terrain above the road toward Sentinel Dome. Cliffs are present a short distance to the north, and steeply sloping rock is present downslope to the east-northeast. Scattered bedrock knobs are present on the slope. (HWA GeoSciences, Inc.,2016)

FIELD EXPLORATION

The field exploration, conducted by **TECHNICON** on October 26 through 30, 2015, consisted of drilling eight (8) exploratory profile borings, installing eight (8) piezometers, and performing eight (8) percolation tests total. Four (4) Piezometers were installed and four (4) percolations tests were conducted at each of the two (2) testing locations. The profile borings were drilled to depth of 12 feet below the existing ground surface (bgs). The purpose of the profile borings was to visually classify and record a continuous log of the native soils encountered and to install the piezometer well casings. The percolation tests were performed adjacent to the piezometers at various depths ranging between 2 and 5 feet bgs to assess the infiltration characteristics of the soil. The locations of the piezometers and percolation tests are indicated on the Site Maps, Figures 2 and 3.

EARTH MATERIALS

The following earth material descriptions are based on the profile borings performed for this study. The eight (8) profile borings (Pz-1 through Pz-8) were drilled using a 2-inch gas powered solid flight auger to a depth of approximately 12 feet bgs. The surface and subsurface soils encountered at each boring location were similar, consisting of silty sand in the upper 1 to 4 feet, underlain by sandy silt, silty sand with gravel, and silty sand comprised of completely weathered in place, friable, decomposed granite. The above is a general description of the earth material profile. A more detailed representation of the stratigraphy at the specific exploration locations is provided on the attached boring logs.

PIEZOMETERS

The eight (8) profile borings were completed with piezometer casings. The piezometers were installed to a depth of 12 feet bgs; with the upper 2 feet consisting of 1-inch diameter non-screened casing and the lower 10 feet consisting of 1-inch diameter screened casing (0.02-inch machine slot size). A sand filter pack was installed between depths of 1.5 and 12 feet bgs and a bentonite seal installed between the surface and 1.5 feet bgs. The piezometer casings generally extended above the surround ground surface to a height of 3 feet for convenience. An approximately 4 to 4.5-foot long, 4-inch diameter Schedule 40 Black PVC standpipe was slipped over the piezometer casing and driven into the ground surface approximately 12 inches for stability. A PVC slip cover was placed over the standpipe.

One piezometer, Pz-3 was installed to a depth of 10.3 feet due to auger refusal within the underlying granite bedrock. Multiple attempts in the area were made to advance the boring to 12 feet; however, the attempts reached refusal shallower than 10.3 feet. The NPS and HWA GeoSciences Inc. was made aware of the refusal and accepted the installation of 10.3 feet.

PERCOLATION TESTS

Eight (8) percolation tests (P-1 through P-8) were performed in borings at depths ranging from 4.5 to 5.25 feet bgs using a 4-inch diameter hand auger and CME 45 truck mounted drill rig utilizing a 7.5-inch hollow stem auger to excavate the test holes. The final diameter of the percolation tests ranged between 4.5 to 8-inches in diameter due to reaming of the test hole that occurred with the augers. Approximately ⅜-inch diameter pea gravel was placed in the bottom 2 inches of the test holes and a 2-inch diameter temporary casing was placed on top of the pea gravel. The annulus of the hole was filled with ⅜-inch diameter pea gravel to prevent caving of the test holes. The location of the percolation tests are shown on the Site Maps, Figures 2 and 3. A summary of the percolation test results is presented in Table 1.

TABLE 1
SUMMARY OF PERCOLATION TESTS

Location	Soil Tested*	Depth (ft)	Percolation Rate (min/in.)	Percolation Rate (gpd/sf)**	Application Rate per County (gpd/sf)**	Depth to Silt or Bedrock**
P-1	Sandy SILT (ML)	4.5	2.0	448.8	1.2	5
P-2	Silty SAND (SM)*	5.1	1.5	598.4	1.2	>12
P-3	Silty SAND (SM)*	4.8	0.9	997.4	NS	10.5
P-4	Poorly Graded Sand with SILT (SP-SM)*	5.25	0.8	1122.1	NS	>12
P-5	Silty SAND (SM)*	5.0	1.0	897.7	NS	>12
P-6	Silty SAND (SM)*	5.0	0.6	1496.1	NS	>12
P-7	Silty SAND (SM)*	5.0	0.4	2244.2	NS	>12
P-8	Silty SAND (SM)*	5.0	0.5	1795.3	NS	>12

*Silty SAND comprised of completely weathered, friable, decomposed granite
**Information provided by HWA GeoSciences, Inc.

CONCLUSIONS AND RECOMMENDATIONS

Glacier Point Existing Drainfield

All four percolation tests (P-1 through P-4) exhibited very high rates, ranging from 0.8 to 2 min/in (30-75 in/hr), with depth to weathered bedrock ("decomposed granite") or low permeability soil ranging from 5 feet to over 12 feet (the maximum boring depth). Although soil percolation rates are too high for treatment, the site is known to experience seasonal high groundwater, typically during spring and early summer. This was the case before and after improvements in 1999 and 2000, and data from 16 monitoring wells indicate groundwater mounding occurs during peak demand times such as weekends and holidays.

Although this site appears favorable based on soil criteria, documented seasonal high ground water may constrain drainfield design, i.e., application rates lower than those based on soil classification or perc tests may be needed. In addition, the Mariposa County Sewage Disposal Rules require a "specially designed system" for soils with measured percolation rates less than 5 min/in. Such designs may include replacing overly permeable soils (i.e., percolation rate less than 1 minute/inch) with "a suitably thick (2 ft) layer of loamy sand or sand."

This area appears suitable for a drainfield, provided 1) a "specially designed system" is used, 2) ground water monitoring indicates favorable ground water levels, and 3) a lower application rate (and resulting larger drainfield) are used, with an adequate reserve area identified to expand the system if needed. (HWA GeoSciences, Inc.,2016)

Glacier Point Old Drainfield

All four percolation tests (P-5 through P-8) exhibited very high rates, ranging from 0.5 to 1 min/in (60-120 in/hr), with depth to bedrock or low permeability soil greater than the maximum boring depth of 12 feet. Although this site appears favorable based on these criteria, the Mariposa County Sewage Disposal Rules require a "specially designed system" for soils with measured percolation rates less than 5 min/in. Such designs may include replacing overly permeable soils (i.e., percolation rate less than 1 minute/inch) with "a suitably thick (2 ft) layer of loamy sand or sand."

This area appears suitable for drainfield, provided 1) a "specially designed system" is used, 2) ground water monitoring indicates favorable ground water levels, and 3) an adequate reserve area is identified to expand the system if needed. (HWA GeoSciences, Inc.,2016)

Ground Water Level

Although no ground water was detected in any of the explorations (conducted in the fall), based on prior studies it is expected to be seasonally high at both Glacier Point sites. The Mariposa County Sewage Disposal Rules require a vertical separation from seasonal high ground water of 5 feet from the bottom of the facility. Assuming a minimum depth of 18 inches, maximum allowable seasonal high ground water is 6.5 feet below grade. Due to this design criterion, ground water level monitoring is recommended in the piezometers for at least one high ground water season, which is expected in late spring or early summer, after the snowmelt. HWA recommends measurements at least once a week. Remote (or all) areas may be monitored using data logging pressure transducers, which can be left in the piezometers for extended time periods and the data retrieved afterwards. (HWA GeoSciences, Inc.,2016)

SUMMARY

Both existing and old drainfield sites appear favorable based on depth of receptor soils and percolation rates, although seasonal high ground water was an issue in the past and needs to be verified. Per the Mariposa County Sewage Disposal Rules, a "specially designed system" may be required due to the overly permeable soils. Such designs may include replacing overly permeable soils (i.e., percolation rate less than 1 minute/inch) with "a suitably thick (2 ft) layer of loamy sand or sand." Alternatively, advanced wastewater treatment systems that do not rely on soil for treatment may be used (see below). Per Mariposa County requirements there shall be a minimum vertical separation of five feet between the bottom of all portions of the trench or bed and any restrictive layer or water table. Unweathered granite bedrock was encountered at a depth of 10.5 feet in one of the piezometer borings (PZ-3), and therefore proposed trenches in the vicinity of that location would have less than 5 feet of vertical separation from the restrictive layer, and thus would not be feasible at those locations. (HWA GeoSciences, Inc.,2016)

Drainfield Design Considerations

If site conditions allow, maximizing the aspect ratio of the drain field (length:width) is always advantageous, i.e., the more geographically spread out the given square feet of absorption area is, the better the system will perform, as the water will be spread over a larger effective area. Orienting the drain fields parallel to slope is also preferred, because shallow ground water flow in the study area likely follows the bedrock surface, which in turn mirrors the topography. (HWA GeoSciences, Inc.,2016)

LIMITATIONS

The information and conclusions presented in this report are based on the information provided regarding the proposed project, and the results of our field and laboratory investigation, combined with interpolation of the subsurface conditions between sample locations. The findings of this study are for the preliminary planning and on-going study of proposed future disposal fields. Additional investigation and testing is necessary to complete design and construction plans and fulfill the requirements of Mariposa County. The nature and extent of the variations may not become evident until further exploration in design stages.

Our professional services were performed, our findings obtained, and our recommendations prepared in accordance with generally accepted engineering principles and practices. This warranty is in lieu of all other warranties either expressed or implied. This report should not be construed as an environmental audit or study.

REFERENCES

Huber, N. King, 1987. *The Geologic Story of Yosemite National Park*, http://www.yosemite.ca.us/library/geologic_story_of_yosemite/maps.html

Huber, N. King, Batemen, Paul C., and Warrhaftig, Karl, 1989. *Geologic Map of Yosemite National Park and Vicinity, California,* USGS Miscellaneous Investigation Series *Map I-1874.*

HWA GeoSciences Inc., 2016. *Hydrogeologic Investigation for Wastewater Improvements, Glacier Point and Crane Flat Campground, June 14, 2016.*

National Park Service, 1995. *Glacier Point Rehabilitation Project Yosemite National Park,* September 7, 1995.

Winzler & Kelly Consulting Engineers, 2006. *Title I Report, Yosemite National Park Glacier Point Wastewater System.* June 2006.

CLOSING

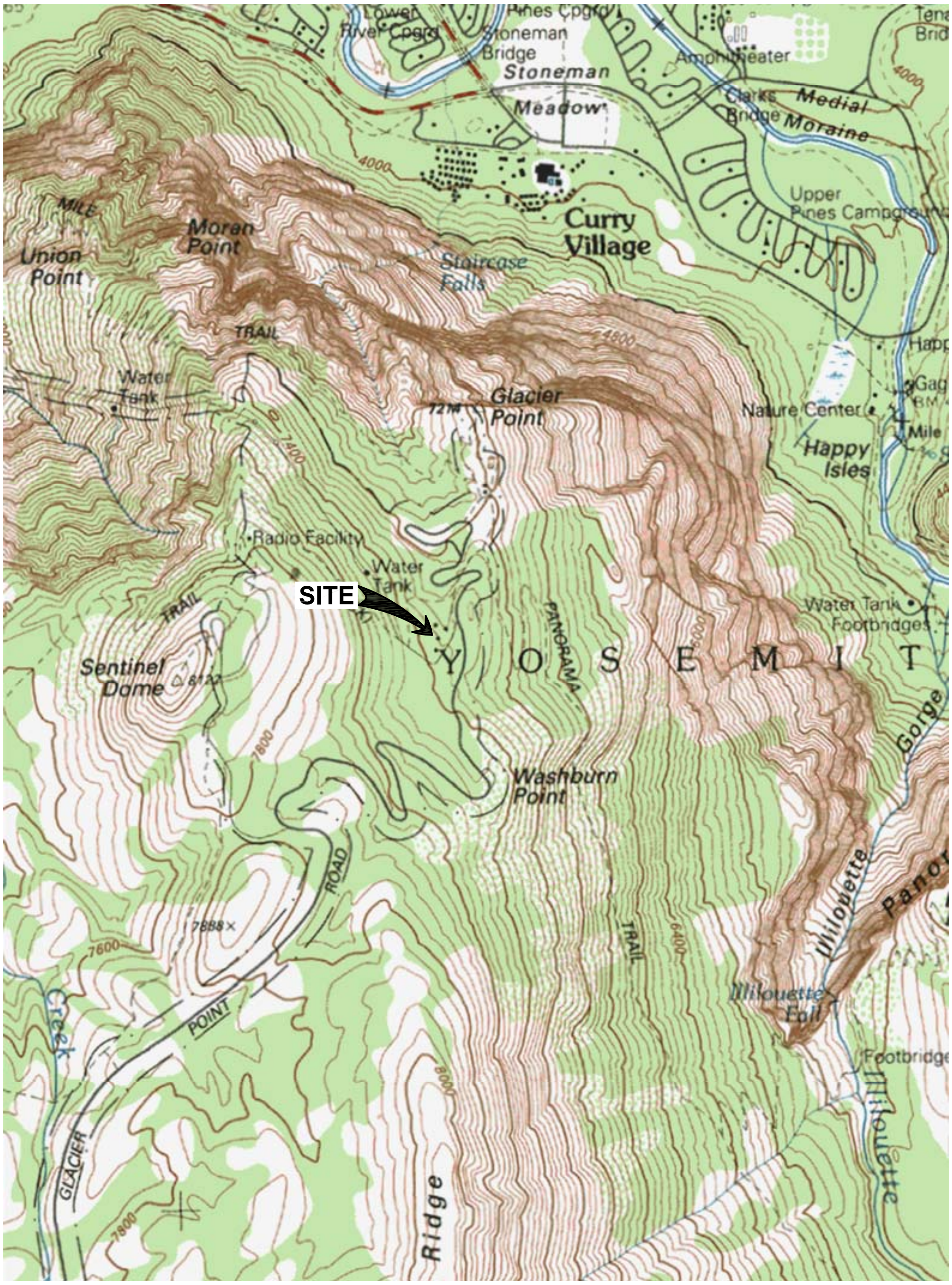
Thank you for your valued business. We trust this report will satisfy your needs. Should you have any questions or if we may be of further assistance to you, please call our office.

Sincerely,
TECHNICON Engineering Services, Inc.

Stephen P. Plauson, PE, GE
Geotechnical Engineering Manager

SPP:AA:mk

Attachments: Vicinity Map, Figure 1
Site Maps, Figures 2 and 3
Key to Symbols and Boring Logs
Percolation Test Reports



N

LAT.: 37.7239°N, LONG.: 119.5750°W, USGS MAP: HALF DOME, DATE: 1992

TECHNICON ENGINEERING SERVICES, INC.	PROJECT: 160577	VICINITY MAP GLACIER POINT YOSEMITE VALLEY, CALIFORNIA	FIGURE 1 NTS
	SOURCE: USGS TOPOGRAPHIC MAPS		

PZ-4 =PIEZOMETER LOCATIONS

P-4 =PERCOLATION TEST LOCATIONS

N

SCALE: 1"=80'

0' 40' 80'

TECHNICON ENGINEERING SERVICES, INC.	PROJECT: 160577	DATE: 11/25/15	SITE MAP GLACIER POINT YOSEMITE VALLEY, CALIFORNIA	FIGURE 2
	SOURCE: GOOGLE EARTH	APPROVED BY: SP		

PZ-8 =PIEZOMETER LOCATIONS
P-8 =PERCOLATION TEST LOCATIONS

TECHNICON ENGINEERING SERVICES, INC.	PROJECT: 160577	DATE: 11/25/15	SITE MAP GLACIER POINT YOSEMITE VALLEY, CALIFORNIA	FIGURE 3
	SOURCE: GOOGLE EARTH	APPROVED BY: SP		

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KEY TO SYMBOLS

PROJECT NAME Glacier Point
DATE OF EXPLORATION 10/26/2015
PROJECT LOCATION Yosemite Valley, CA
PROJECT NUMBER 160577

LITHOLOGIC SYMBOLS (Unified Soil Classification System)

	FILL
SW	WELL GRADED SAND
SP	POORLY GRADED SAND
SM	SILTY SAND
SC	CLAYEY SAND
PT	PEAT
OL	LOW PLASTICITY ORGANIC SILT
OH	HIGH PLASTICITY ORGANIC SILT
ML	LOW PLASTICITY SILT
MH	HIGH PLASTICITY SILT
GW	WELL GRADED GRAVEL
GP	POORLY GRADED GRAVEL
GM	SILTY GRAVEL
GC	CLAYEY GRAVEL
CL	LOW PLASTICITY CLAY
CH	HIGH PLASTICITY CLAY

SAMPLER SYMBOLS

- STANDARD PENETRATION TEST
- CALIFORNIA SAMPLER
- MODIFIED CALIFORNIA SAMPLER
- SHELBY TUBE SAMPLER
- ROCK CORE BARREL
- BULK SAMPLE

- ∇ Water Level at Time of Drilling
- ▼ Water Level at End of Drilling
- ▼ Water Level After 24 Hours
- – – Assumed stratum line
- —— Observed stratum line

Note 1: The degree of saturation shown on the boring logs is based on an assumed specific gravity of 2.65. The actual degree of saturation may vary.

Note 2: The stratum lines shown on the logs represent the approximate boundary between soil types; the actual in-situ transition may be gradual.

ABBREVIATIONS

LL	- LIQUID LIMIT (%)
PI	- PLASTIC INDEX (%)
W	- MOISTURE CONTENT (%)
DD	- DRY DENSITY (PCF)
S	- DEGREE OF SATURATION (%)
NP	- NON PLASTIC
-200	PERCENT PASSING NO. 200 SIEVE
PP	- POCKET PENETROMETER (TSF)
TV	- TORVANE
PID	- PHOTOIONIZATION DETECTOR
UC	- UNCONFINED COMPRESSION
ppm	- PARTS PER MILLION

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BORING P-1

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/26/15 **COMPLETED** 10/26/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE
BORING DEPTH 4.5 ft
DRILLING METHOD Hand Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, moist, fine to medium grained Light brown, fine to coarse grained, trace fine gravel				

NOTES:
1. Bottom of boring at 4.5 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/26/15.

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BORING P-2

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/26/15 **COMPLETED** 10/26/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE
BORING DEPTH 5.1 ft
DRILLING METHOD Hand Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, moist, fine to coarse grained				
				Silty SAND (SM) - light brown, moist, fine to coarse grained, comprised of completely weathered friable decomposed granite				
5								

NOTES:
1. Bottom of boring at 5.1 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/26/15.

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BORING P-3

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/26/15 **COMPLETED** 10/26/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE
BORING DEPTH 4.8 ft
DRILLING METHOD Hand Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, moist, fine to coarse grained				
				Silty SAND (SM) - light gray, moist, fine to coarse grained, comprised of completely weathered friable decomposed granite				

NOTES:
1. Bottom of boring at 4.8 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/26/15.

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BORING P-4

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/26/15 **COMPLETED** 10/26/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE
BORING DEPTH 5.25 ft
DRILLING METHOD Hand Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, moist, fine to medium grained				
				Silty SAND (SM) - light brown, moist, fine to coarse grained, comprised of completely weathered friable decomposed granite				
5				**Poorly Graded SAND (SP-SM)** - light grayish brown, moist, fine to coarse grained, with silt, comprised of completely weathered friable decomposed granite				

NOTES:
1. Bottom of boring at 5.3 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/26/15.

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BORING P-5

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/27/15 **COMPLETED** 10/27/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE
BORING DEPTH 5 ft
DRILLING METHOD Hand Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, moist, fine to medium grained				
				Silty SAND (SM) - light brown, moist, fine to coarse grained, comprised of completely weathered friable decomposed granite				
5								

NOTES:
1. Bottom of boring at 5.0 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/27/15.

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BORING P-6

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/27/15 **COMPLETED** 10/27/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE
BORING DEPTH 5 ft
DRILLING METHOD Hand Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - dark brown, moist, fine to coarse grained, trace fine gravel Grayish brown, moist, fine grained, with fine to coarse gravel Fine to coarse grained				
5				**Silty SAND (SM)** - yellowish brown, moist, fine to coarse grained, trace fine gravel, comprised of completely weathered friable decomposed granite				

NOTES:
1. Bottom of boring at 5.0 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/27/15.

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BORING P-7

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/27/15 **COMPLETED** 10/27/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE
BORING DEPTH 5 ft
DRILLING METHOD Hand Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - dark brown, moist, fine to coarse grained, trace fine gravel Brown, with fine to coarse gravel, comprised of completely weathered friable decomposed granite Increased coarse gravel				
5								

NOTES:
1. Bottom of boring at 5.0 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/27/15.

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Fax: (559) 276-9344

BORING P-8

PROJECT NAME	Glacier Point	**PROJECT NUMBER**	160577
PROJECT LOCATION	Yosemite Valley, CA	**SURFACE DESCRIPTION**	Bare Soil
DATE STARTED	10/27/15 **COMPLETED** 10/27/15	**GROUND ELEVATION**	
DRILLING CONTRACTOR	TECHNICON Engineering Services	**GROUND WATER LEVEL**	No groundwater encountered.
DRILL RIG TYPE		**BORING DEPTH**	5 ft
DRILLING METHOD	Hand Auger	**LOGGED BY**	A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, moist, fine to coarse grained, trace fine gravel				
				Silty SAND (SM) - brown, moist, fine to coarse grained, trace fine to coarse gravel, comprised of completely weathered friable decomposed granite Increased gravel				
5				Yellowish brown, trace fine gravel, increased silt				

NOTES:
1. Bottom of boring at 5.0 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/27/15.

Technicon Engineering Services, Inc
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Fresno, CA 93622
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Fax: (559) 276-9344

BORING Pz-1

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/26/15 **COMPLETED** 10/26/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE CME 45
BORING DEPTH 12 ft
DRILLING METHOD 4-inch Solid Flight Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, moist, fine to medium grained				
				Fine to coarse grained, trace fine gravel				
5				Light brown, fine grained, no gravel				
				Sandy SILT (ML) - light grayish brown, moist, with fine sand				
				Light gray				
10								

NOTES:
1. Bottom of boring at 12.0 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/26/15.

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BORING Pz-2

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/28/15 **COMPLETED** 10/28/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE Hand Auger
BORING DEPTH 12 ft
DRILLING METHOD 2-inch Solid Flight Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, fine to medium grained				
				Silty SAND (SM) - light brown, fine to coarse grained, comprised of completely weathered friable decomposed granite				
5								
10								

NOTES:
1. Bottom of boring at 12.0 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/28/15.

BOREHOLE - TECHNICON.GDT - 7/6/16 15:10 - Z:\TESDATA\USERS\ADAM A\YOSEMITE\150783 - GLACIER POINT\CALCS\GINT.GPJ

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BORING Pz-3

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/28/15 **COMPLETED** 10/28/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE Hand Auger
BORING DEPTH 10.25 ft
DRILLING METHOD 2-inch Solid Flight Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, fine to medium grained				
				Silty SAND (SM) - light brown, fine to coarse grained, comprised of completely weathered friable decomposed granite				
5				**Decomposed GRANITE** - light gray, drilling returns comprised of fine rock dust				
10								

NOTES:
1. Bottom of boring at 10.3 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/28/15.

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BORING Pz-4

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/28/15 **COMPLETED** 10/28/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE Hand Auger
BORING DEPTH 12 ft
DRILLING METHOD 2-inch Solid Flight Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, moist, fine to medium grained				
				Silty SAND (SM) - light brown, moist, fine to coarse grained, comprised of completely weathered friable decomposed granite				
5								
10								

NOTES:
1. Bottom of boring at 12.0 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/28/15.

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BORING Pz-5

PROJECT NAME Glacier Point	**PROJECT NUMBER** 160577
PROJECT LOCATION Yosemite Valley, CA	**SURFACE DESCRIPTION** Bare Soil
DATE STARTED 10/27/15 **COMPLETED** 10/27/15	**GROUND ELEVATION**
DRILLING CONTRACTOR TECHNICON Engineering Services	**GROUND WATER LEVEL** No groundwater encountered.
DRILL RIG TYPE Hand Auger	**BORING DEPTH** 12 ft
DRILLING METHOD 2-inch Solid Flight Auger	**LOGGED BY** A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, moist, fine to medium grained				
				Silty SAND (SM) - light brown, moist, fine to coarse grained, comprised of completely weathered friable decomposed granite				
5								
10								

NOTES:
1. Bottom of boring at 12.0 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/27/15.

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BORING Pz-6

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/27/15 **COMPLETED** 10/27/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE Hand Auger
BORING DEPTH 12 ft
DRILLING METHOD 2-inch Solid Flight Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, moist, fine to medium grained				
5				Gray, with fine to coarse gravel				
10				**Silty SAND (SM)** - light brown, moist, fine to medium grained, comprised of completely weathered friable decomposed granite				

NOTES:
1. Bottom of boring at 12.0 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/27/15.

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BORING Pz-7

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/27/15 **COMPLETED** 10/27/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE Hand Auger
BORING DEPTH 12 ft
DRILLING METHOD 2-inch Solid Flight Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, moist, fine to medium grained				
				Silty SAND (SM) - light brown, moist, fine to medium grained, with fine gravel, comprised of completely weathered friable decomposed granite				
5								
				No gravel				
10								

NOTES:
1. Bottom of boring at 12.0 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/27/15.

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BORING Pz-8

PROJECT NAME Glacier Point
PROJECT NUMBER 160577
PROJECT LOCATION Yosemite Valley, CA
SURFACE DESCRIPTION Bare Soil
DATE STARTED 10/27/15 **COMPLETED** 10/27/15
GROUND ELEVATION
DRILLING CONTRACTOR TECHNICON Engineering Services
GROUND WATER LEVEL No groundwater encountered.
DRILL RIG TYPE Hand Auger
BORING DEPTH 12 ft
DRILLING METHOD 2-inch Solid Flight Auger
LOGGED BY A. AhTye **CHECKED BY** S. Plauson

DEPTH (ft)	SAMPLE TYPE	BLOWS/ft	GRAPHIC LOG	MATERIAL DESCRIPTION	DRY DENSITY (pcf)	MOISTURE (%)	OTHER TESTS	REMARKS
0				**Silty SAND (SM)** - brown, moist, fine to medium grained				
				Silty SAND (SM) - light brown, moist, fine to coarse grained, comprised of completely weathered friable decomposed granite				
5				With fine gravel				
				No fine gravel				
10								

NOTES:
1. Bottom of boring at 12.0 feet.
2. No groundwater encountered.
3. Boring backfilled with soil cuttings 10/27/15.

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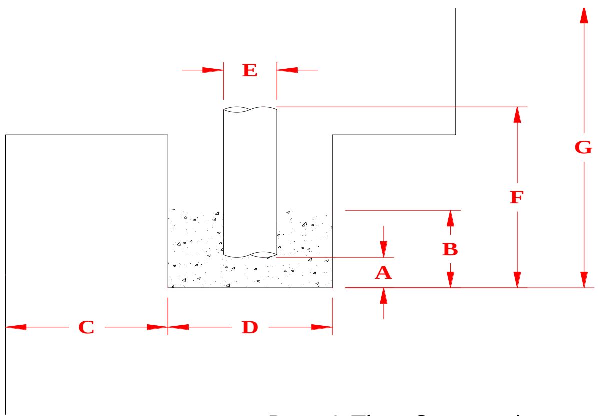
PERCOLATION TEST DATA SHEET

Project Name:	Glacier Point	Project No.:	160577
Project Location:	Yosemite Valley, CA	Pit No.:	P-1



A.	Gravel Layer Depth, in.	2
B.	Total Gravel Thickness, in.	14
C.	Distance from Shelf, ft.	-
D.	Hole Diameter, in.	4.5 - 6.0
E.	Casing Diameter, in.	2
F.	Reference Depth, in.	93
G	Hole Depth, ft. (in.)	4.5 (54)
	Depth to Groundwater	-
Soil Type	Sandy SILT (ML)	

Date & Time Saturated 10/26/15 12:00 PM

Depth of Water after 24-hour Saturation 0

Begin Test	Initial Depth to Water*, in.	Refilled	End Test	Final Depth to Water*, in.	Test Duration, min.	Water Drop, in.	**Drop Rate min./in.****
1:00	85.0	X	13:02	89.0	2.3	4.0	**0.9**
13:02	85.0	X	13:05	89.0	3.2	4.0	**1.3**
13:05	85.0	X	13:09	89.0	3.6	4.0	**1.4**
13:09	85.0	X	13:13	89.0	4.1	4.0	**1.6**
13:13	85.0	X	13:17	89.0	4.5	4.0	**1.8**
13:17	85.0	X	13:22	89.0	4.6	4.0	**1.8**
13:22	85.0	X	13:26	89.0	4.6	4.0	**1.8**
13:26	85.0	X	13:31	89.0	4.7	4.0	**1.9**
13:31	85.0	X	13:36	89.0	4.9	4.0	**2.0**
13:36	85.0	X	13:41	89.0	5.0	4.0	**2.0**
13:41	85.0	X	13:46	89.0	4.9	4.0	**2.0**
13:46	85.0	X	13:51	89.0	4.9	4.0	**2.0**
13:51	85.0	X	13:56	89.0	5.0	4.0	**2.0**
13:56	85.0	X	14:01	89.0	5.0	4.0	**2.0**
						Average (Last 6) =	**2.0**

[1]Silty Sand comprised of completely weathered friable decomposed granite

*Depth below reference datum

**Corrected for full depth gravel in annulus

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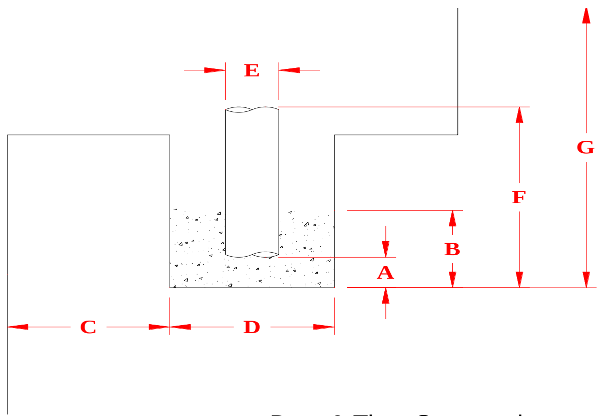
PERCOLATION TEST DATA SHEET

Project Name:	Glacier Point	Project No.:	160577
Project Location:	Yosemite Valley, CA	Pit No.:	P-2



A.	Gravel Layer Depth, in.	2
B.	Total Gravel Thickness, in.	14
C.	Distance from Shelf, ft.	-
D.	Hole Diameter, in.	4.5 - 6.0
E.	Casing Diameter, in.	2
F.	Reference Depth, in.	98
G	Hole Depth, ft. (in.)	5.1 (61)
	Depth to Groundwater	-
	Soil Type	Silty SAND (SM)[1]

Date & Time Saturated 10/26/15 1:00 PM
Depth of Water after 24-hour Saturation 0

Begin Test	Initial Depth to Water*, in.	Refilled	End Test	Final Depth to Water*, in.	Test Duration, min.	Water Drop, in.	**Drop Rate min./in.****
13:00	90.0	X	13:02	94.0	2.6	4.0	**1.0**
13:02	90.0	X	13:05	94.0	2.6	4.0	**1.0**
13:05	90.0	X	13:08	94.0	3.1	4.0	**1.2**
13:08	90.0	X	13:11	94.0	3.1	4.0	**1.2**
13:11	90.0	X	13:14	94.0	3.3	4.0	**1.3**
13:14	90.0	X	13:17	94.0	3.4	4.0	**1.4**
13:17	90.0	X	13:21	94.0	3.5	4.0	**1.4**
13:21	90.0	X	13:24	94.0	3.6	4.0	**1.4**
13:24	90.0	X	13:28	94.0	3.6	4.0	**1.5**
13:28	90.0	X	13:31	94.0	3.6	4.0	**1.4**
13:31	90.0	X	13:35	94.0	3.7	4.0	**1.5**
13:35	90.0	X	13:38	94.0	3.6	4.0	**1.5**
13:38	90.0	X	13:42	94.0	3.8	4.0	**1.5**
13:42	90.0	X	13:46	94.0	3.8	4.0	**1.5**
13:46	90.0	X	13:50	94.0	3.8	4.0	**1.5**
13:50	90.0	X	13:54	94.0	3.8	4.0	**1.5**
13:54	90.0	X	13:58	94.0	3.8	4.0	**1.5**
13:58	90.0	X	14:02	94.0	3.8	4.0	**1.5**

Average (Last 6) = **1.5**

[1]Silty Sand comprised of completely weathered friable decomposed granite
*Depth below reference datum
**Corrected for full depth gravel in annulus

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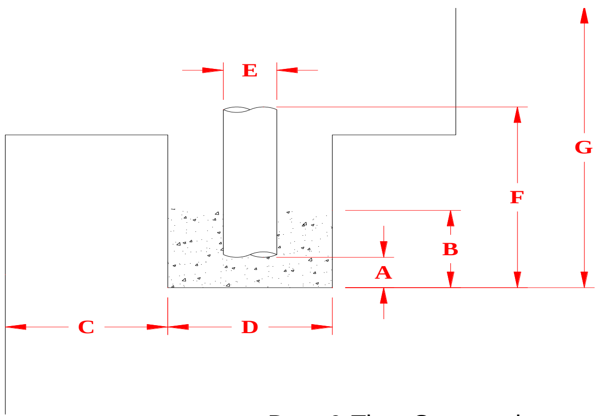
PERCOLATION TEST DATA SHEET

Project Name:	Glacier Point	Project No.:	160577
Project Location:	Yosemite Valley, CA	Pit No.:	P-3



A.	Gravel Layer Depth, in.	2
B.	Total Gravel Thickness, in.	14
C.	Distance from Shelf, ft.	-
D.	Hole Diameter, in.	4.5 - 6.0
E.	Casing Diameter, in.	2
F.	Reference Depth, in.	104
G	Hole Depth, ft. (in.)	4.8 (58)
	Depth to Groundwater	-
	Soil Type	Silty SAND (SM)[1]

Date & Time Saturated 10/28/15 2:30 PM

Depth of Water after 24-hour Saturation 0

Begin Test	Initial Depth to Water*, in.	Refilled	End Test	Final Depth to Water*, in.	Test Duration, min.	Water Drop, in.	**Drop Rate min./in.****
14:30	96.00	X	14:34	100.0	4.1	4.0	**1.6**
14:34	96.00	X	14:37	100.0	3.4	4.0	**1.4**
14:37	96.00	X	14:40	100.0	2.4	4.0	**1.0**
14:40	96.00	X	14:42	100.0	2.1	4.0	**0.9**
14:42	96.00	X	14:44	100.0	2.1	4.0	**0.8**
14:44	96.00	X	14:46	100.0	2.2	4.0	**0.9**
14:46	96.00	X	14:48	100.0	2.2	4.0	**0.9**
14:48	96.00	X	14:51	100.0	2.3	4.0	**0.9**
14:51	96.00	X	14:53	100.0	2.3	4.0	**0.9**
14:53	96.00	X	14:55	100.0	2.3	4.0	**0.9**
14:55	96.00	X	14:58	100.0	2.2	4.0	**0.9**
14:58	96.00	X	15:00	100.0	2.5	4.0	**1.0**
15:00	96.00	X	15:02	100.0	2.4	4.0	**0.9**
15:02	96.00	X	15:05	100.0	2.3	4.0	**0.9**
15:05	96.00	X	15:07	100.0	2.3	4.0	**0.9**
15:07	96.00	X	15:09	100.0	2.3	4.0	**0.9**
15:09	96.00	X	15:11	100.0	2.3	4.0	**0.9**
15:11	96.00	X	15:14	100.0	2.3	4.0	**0.9**
15:14	96.00	X	15:16	100.0	2.3	4.0	**0.9**
15:16	96.00	X	15:18	100.0	2.3	4.0	**0.9**
15:18	96.00	X	15:20	100.0	2.3	4.0	**0.9**
15:20	96.00	X	15:23	100.0	2.3	4.0	**0.9**
15:23	96.00	X	15:25	100.0	2.3	4.0	**0.9**
						Average (Last 6) =	**0.9**

[1]Silty Sand comprised of completely weathered friable decomposed granite

*Depth below reference datum

**Corrected for full depth gravel in annulus

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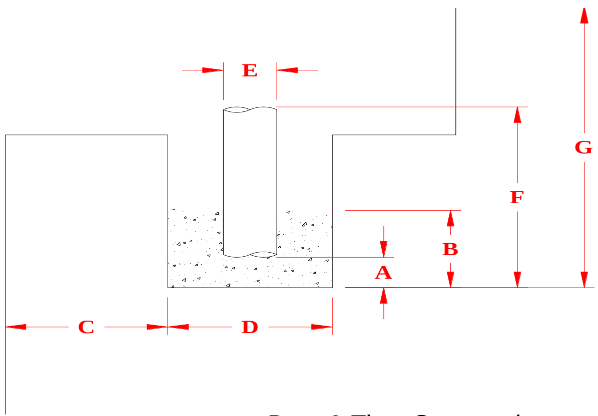
PERCOLATION TEST DATA SHEET

Project Name:	Glacier Point	Project No.:	160577
Project Location:	Yosemite Valley, CA	Pit No.:	P-4 pg 1 of 2



A.	Gravel Layer Depth, in.	4
B.	Total Gravel Thickness, in.	16
C.	Distance from Shelf, ft.	-
D.	Hole Diameter, in.	4.5 - 6.0
E.	Casing Diameter, in.	2
F.	Reference Depth, in.	96
G	Hole Depth, ft. (in.)	5.25 (63)
	Depth to Groundwater	-
	Soil Type	Poorly Graded Sand with Silt (SP-SM)[1]

Date & Time Saturated 10/30/15 2:30 PM
Depth of Water after 24-hour Saturation 0

Begin Test	Initial Depth to Water*, in.	Refilled	End Test	Final Depth to Water*, in.	Test Duration, min.	Water Drop, in.	**Drop Rate min./in.****
14:30	88.0	X	14:31	92.0	1.1	4.0	**0.4**
14:31	88.0	X	14:32	92.0	1.3	4.0	**0.5**
14:32	88.0	X	14:33	92.0	1.3	4.0	**0.5**
14:33	88.0	X	14:34	92.0	1.4	4.0	**0.6**
14:34	88.0	X	14:35	92.0	1.5	4.0	**0.6**
14:35	88.0	X	14:36	92.0	1.6	4.0	**0.7**
14:36	88.0	X	14:37	92.0	1.7	4.0	**0.7**
14:37	88.0	X	14:39	92.0	1.8	4.0	**0.7**
14:39	88.0	X	14:41	92.0	1.9	4.0	**0.7**
14:41	88.0	X	14:43	92.0	2.0	4.0	**0.8**
14:43	88.0	X	14:45	92.0	2.0	4.0	**0.8**
14:45	88.0	X	14:47	92.0	1.9	4.0	**0.8**
14:47	88.0	X	14:49	92.0	1.9	4.0	**0.8**
14:49	88.0	X	14:51	92.0	1.9	4.0	**0.8**
14:51	88.0	X	14:53	92.0	2.0	4.0	**0.8**
14:53	88.0	X	14:55	92.0	2.0	4.0	**0.8**
14:55	88.0	X	14:57	92.0	2.0	4.0	**0.8**
14:57	88.0	X	14:59	92.0	2.0	4.0	**0.8**
14:59	88.0	X	15:01	92.0	2.0	4.0	**0.8**
15:01	88.0	X	15:03	92.0	2.0	4.0	**0.8**
15:03	88.0	X	15:05	92.0	2.0	4.0	**0.8**
15:05	88.0	X	15:07	92.0	2.0	4.0	**0.8**
15:07	88.0	X	15:09	92.0	2.0	4.0	**0.8**

See Next Page

[1]Silty Sand comprised of completely weathered friable decomposed granite
*Depth below reference datum
**Corrected for full depth gravel in annulus

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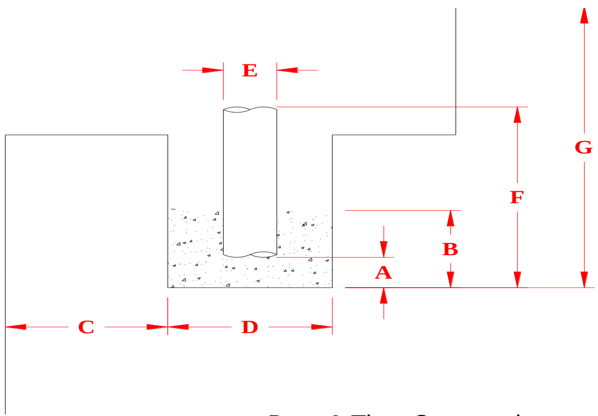
PERCOLATION TEST DATA SHEET

Project Name:	Glacier Point	Project No.:	160577
Project Location:	Yosemite Valley, CA	Pit No.:	P-4 pg 2 of 2



A.	Gravel Layer Depth, in.	4
B.	Total Gravel Thickness, in.	16
C.	Distance from Shelf, ft.	-
D.	Hole Diameter, in.	4.5 - 6.0
E.	Casing Diameter, in.	2
F.	Reference Depth, in.	96
G	Hole Depth, ft. (in.)	5.25 (63)
	Depth to Groundwater	-
	Soil Type	Poorly Graded Sand with Silt (SP-SM)[1]

Date & Time Saturated 10/30/15 2:30 PM
Depth of Water after 24-hour Saturation 0

Begin Test	Initial Depth to Water*, in.	Refilled	End Test	Final Depth to Water*, in.	Test Duration, min.	Water Drop, in.	**Drop Rate min./in.****
15:09	88.0	X	15:11	92.0	2.0	4.0	**0.8**
15:11	88.0	X	15:13	92.0	2.0	4.0	**0.8**
15:13	88.0	X	15:15	92.0	2.0	4.0	**0.8**
15:15	88.0	X	15:17	92.0	2.0	4.0	**0.8**
15:17	88.0	X	15:19	92.0	2.0	4.0	**0.8**
15:19	88.0	X	15:21	92.0	2.0	4.0	**0.8**
15:21	88.0	X	15:23	92.0	2.0	4.0	**0.8**
15:23	88.0	X	15:25	92.0	2.0	4.0	**0.8**
15:25	88.0	X	15:27	92.0	2.0	4.0	**0.8**
15:27	88.0	X	15:29	92.0	2.0	4.0	**0.8**
15:29	88.0	X	15:31	92.0	2.0	4.0	**0.8**
15:31	88.0	X	15:33	92.0	2.0	4.0	**0.8**

Average (Last 6) = **0.8**

[1]Silty Sand comprised of completely weathered friable decomposed granite
*Depth below reference datum
**Corrected for full depth gravel in annulus

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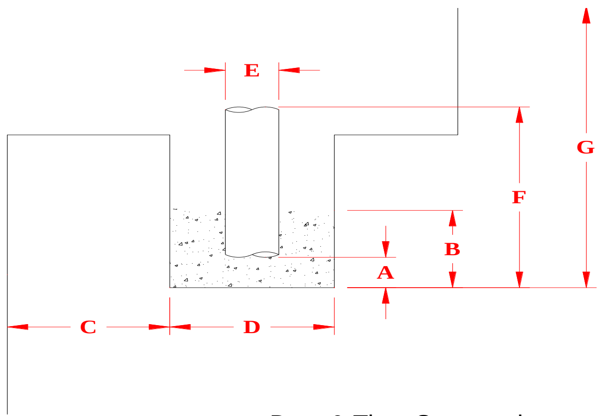
Construction Testing & Inspection * *Geotechnical & Environmental Engineering*

PERCOLATION TEST DATA SHEET

Project Name:	Glacier Point	Project No.:	160577
Project Location:	Yosemite Valley, CA	Pit No.:	P-5



A.	Gravel Layer Depth, in.	2
B.	Total Gravel Thickness, in.	15
C.	Distance from Shelf, ft.	-
D.	Hole Diameter, in.	4.5 - 6.0
E.	Casing Diameter, in.	2
F.	Reference Depth, in.	117
G	Hole Depth, ft. (in.)	5 (60)
	Depth to Groundwater	-
Soil Type	Silty SAND (SM)[1]	

Date & Time Saturated 10/29/15 9:00 AM

Depth of Water after 24-hour Saturation 0

Begin Test	Initial Depth to Water*, in.	Refilled	End Test	Final Depth to Water*, in.	Test Duration, min.	Water Drop, in.	**Drop Rate min./in.****
9:00	109.0	X	9:07	113.0	7.2	4.0	**2.9**
9:07	109.0	X	9:14	113.0	7.2	4.0	**2.9**
9:14	109.0	X	9:16	113.0	2.4	4.0	**1.0**
9:16	109.0	X	9:18	113.0	1.9	4.0	**0.7**
9:18	109.0	X	9:20	113.0	2.0	4.0	**0.8**
9:20	109.0	X	9:22	113.0	2.0	4.0	**0.8**
9:22	109.0	X	9:24	113.0	2.0	4.0	**0.8**
9:24	109.0	X	9:26	113.0	2.0	4.0	**0.8**
9:26	109.0	X	9:28	113.0	2.2	4.0	**0.9**
9:28	109.0	X	9:31	113.0	2.4	4.0	**1.0**
9:31	109.0	X	9:33	113.0	2.3	4.0	**0.9**
9:33	109.0	X	9:36	113.0	2.4	4.0	**0.9**
9:36	109.0	X	9:38	113.0	2.3	4.0	**0.9**
9:38	109.0	X	9:40	113.0	2.4	4.0	**0.9**
9:40	109.0	X	9:43	113.0	2.3	4.0	**0.9**
9:43	109.0	X	9:45	113.0	2.4	4.0	**1.0**
9:45	109.0	X	9:47	113.0	2.4	4.0	**1.0**
9:47	109.0	X	9:50	113.0	2.3	4.0	**0.9**
9:50	109.0	X	9:52	113.0	2.4	4.0	**1.0**
9:52	109.0	X	9:54	113.0	2.4	4.0	**1.0**
9:54	109.0	X	9:56	113.0	2.4	4.0	**1.0**
9:56	109.0	X	9:59	113.0	2.4	4.0	**1.0**
9:59	109.0	X	10:01	113.0	2.4	4.0	**1.0**
						Average (Last 6) =	**1.0**

[1]Silty Sand comprised of completely weathered friable decomposed granite

*Depth below reference datum

**Corrected for full depth gravel in annulus

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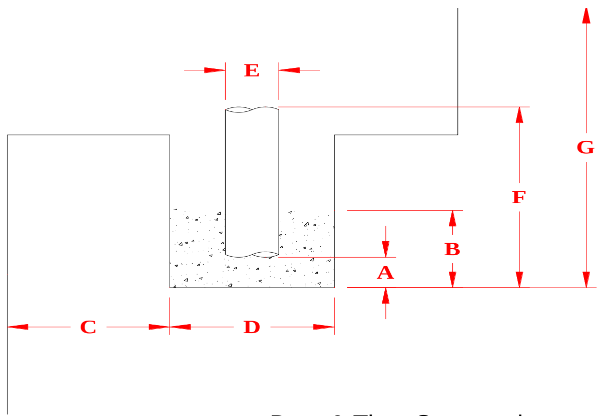
PERCOLATION TEST DATA SHEET

Project Name:	Glacier Point	Project No.:	160577
Project Location:	Yosemite Valley, CA	Pit No.:	P-6 pg 1 of 2



A.	Gravel Layer Depth, in.	4
B.	Total Gravel Thickness, in.	13
C.	Distance from Shelf, ft.	-
D.	Hole Diameter, in.	4.5 - 6.0
E.	Casing Diameter, in.	2
F.	Reference Depth, in.	61
G	Hole Depth, ft. (in.)	5 (60)
	Depth to Groundwater	-
Soil Type	Silty SAND (SM)[1]	

Date & Time Saturated 10/29/15 9:00 AM
Depth of Water after 24-hour Saturation 0

Begin Test	Initial Depth to Water*, in.	Refilled	End Test	Final Depth to Water*, in.	Test Duration, min.	Water Drop, in.	**Drop Rate min./in.***
9:00	51.8	X	9:01	55.8	0.8	4.0	**0.3**
9:01	51.8	X	9:03	55.8	1.4	4.0	**0.5**
9:03	51.8	X	9:05	55.8	1.3	4.0	**0.5**
9:05	51.8	X	9:07	55.8	1.4	4.0	**0.6**
9:07	51.8	X	9:09	55.8	1.5	4.0	**0.6**
9:09	51.8	X	9:11	55.8	1.4	4.0	**0.6**
9:11	51.8	X	9:13	55.8	1.5	4.0	**0.6**
9:13	51.8	X	9:15	55.8	1.6	4.0	**0.6**
9:15	51.8	X	9:17	55.8	1.6	4.0	**0.7**
9:17	51.8	X	9:19	55.8	1.7	4.0	**0.7**
9:19	51.8	X	9:21	55.8	1.6	4.0	**0.7**
9:21	51.8	X	9:23	55.8	1.7	4.0	**0.7**
9:23	51.8	X	9:25	55.8	1.6	4.0	**0.7**
9:25	51.8	X	9:27	55.8	1.6	4.0	**0.7**
9:27	51.8	X	9:29	55.8	1.6	4.0	**0.7**
9:29	51.8	X	9:31	55.8	1.6	4.0	**0.7**
9:31	51.8	X	9:33	55.8	1.6	4.0	**0.7**
9:33	51.8	X	9:35	55.8	1.6	4.0	**0.7**
9:35	51.8	X	9:37	55.8	1.6	4.0	**0.7**
9:37	51.8	X	9:39	55.8	1.6	4.0	**0.7**
9:39	51.8	X	9:41	55.8	1.6	4.0	**0.6**
9:41	51.8	X	9:43	55.8	1.6	4.0	**0.6**
9:43	51.8	X	9:45	55.8	1.6	4.0	**0.6**

See Next Page

[1]Silty Sand comprised of completely weathered friable decomposed granite
*Depth below reference datum
**Corrected for full depth gravel in annulus

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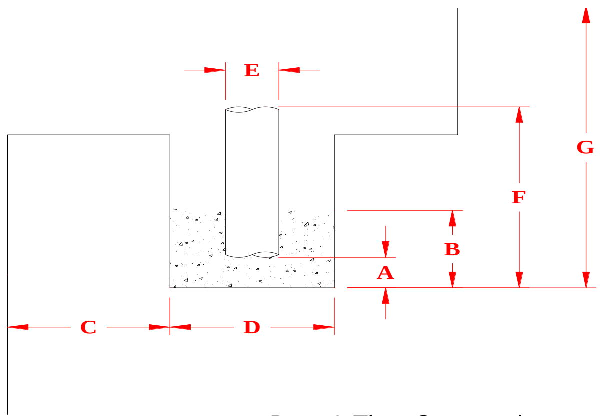
PERCOLATION TEST DATA SHEET

Project Name:	Glacier Point	Project No.:	160577
Project Location:	Yosemite Valley, CA	Pit No.:	P-6 pg 2 of 2



A.	Gravel Layer Depth, in.	4
B.	Total Gravel Thickness, in.	13
C.	Distance from Shelf, ft.	-
D.	Hole Diameter, in.	4.5 - 6.0
E.	Casing Diameter, in.	2
F.	Reference Depth, in.	61
G	Hole Depth, ft. (in.)	5 (60)
	Depth to Groundwater	-
Soil Type	Silty SAND (SM)[1]	

Date & Time Saturated 10/29/15 9:00 AM
Depth of Water after 24-hour Saturation 0

Begin Test	Initial Depth to Water*, in.	Refilled	End Test	Final Depth to Water*, in.	Test Duration, min.	Water Drop, in.	**Drop Rate min./in.** **
9:45	51.8	X	9:47	55.8	1.6	4.0	**0.6**
9:47	51.8	X	9:49	55.8	1.6	4.0	**0.6**
9:49	51.8	X	9:51	55.8	1.6	4.0	**0.6**
9:51	51.8	X	9:53	55.8	1.6	4.0	**0.6**
9:53	51.8	X	9:55	55.8	1.6	4.0	**0.6**
9:55	51.8	X	9:57	55.8	1.6	4.0	**0.6**
9:57	51.8	X	9:59	55.8	1.6	4.0	**0.6**
9:59	51.8	X	10:01	55.8	1.6	4.0	**0.6**
10:01	51.8	X	10:03	55.8	1.6	4.0	**0.6**

Average (Last 6) = 0.6

[1]Silty Sand comprised of completely weathered friable decomposed granite
*Depth below reference datum
**Corrected for full depth gravel in annulus

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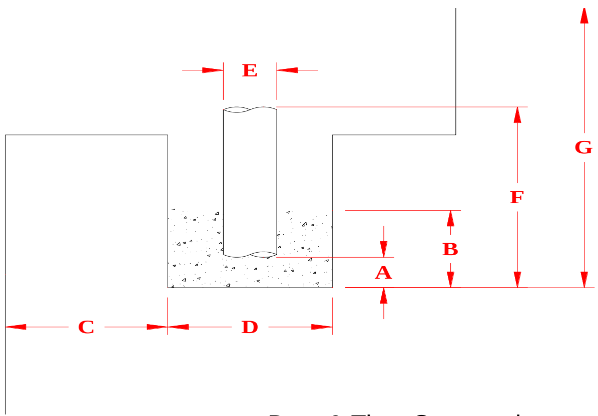
PERCOLATION TEST DATA SHEET

Project Name:	Glacier Point	Project No.:	160577
Project Location:	Yosemite Valley, CA	Pit No.:	P-7 pg 1 of 2



A.	Gravel Layer Depth, in.	2
B.	Total Gravel Thickness, in.	12
C.	Distance from Shelf, ft.	-
D.	Hole Diameter, in.	4.5 - 6.0
E.	Casing Diameter, in.	2
F.	Reference Depth, in.	98
G	Hole Depth, ft. (in.)	5 (60)
	Depth to Groundwater	-
Soil Type		Silty SAND (SM)[1]

Date & Time Saturated 10/29/15 10:30 AM
Depth of Water after 24-hour Saturation 0

Begin Test	Initial Depth to Water*, in.	Refilled	End Test	Final Depth to Water*, in.	Test Duration, min.	Water Drop, in.	**Drop Rate min./in.****
10:30	90.0	X	10:31	94.0	0.8	4.0	**0.3**
10:31	90.0	X	10:32	94.0	0.9	4.0	**0.4**
10:32	90.0	X	10:33	94.0	1.0	4.0	**0.4**
10:33	90.0	X	10:34	94.0	1.1	4.0	**0.4**
10:34	90.0	X	10:35	94.0	1.0	4.0	**0.4**
10:35	90.0	X	10:36	94.0	1.1	4.0	**0.4**
10:36	90.0	X	10:37	94.0	1.0	4.0	**0.4**
10:37	90.0	X	10:38	94.0	1.1	4.0	**0.4**
10:38	90.0	X	10:39	94.0	1.1	4.0	**0.4**
10:39	90.0	X	10:40	94.0	1.0	4.0	**0.4**
10:40	90.0	X	10:41	94.0	1.0	4.0	**0.4**
10:41	90.0	X	10:42	94.0	1.0	4.0	**0.4**
10:42	90.0	X	10:43	94.0	1.0	4.0	**0.4**
10:43	90.0	X	10:44	94.0	1.0	4.0	**0.4**
10:44	90.0	X	10:45	94.0	1.0	4.0	**0.4**
10:45	90.0	X	10:46	94.0	1.0	4.0	**0.4**
10:46	90.0	X	10:47	94.0	1.0	4.0	**0.4**
10:47	90.0	X	10:48	94.0	1.0	4.0	**0.4**
10:48	90.0	X	10:49	94.0	1.0	4.0	**0.4**
10:49	90.0	X	10:50	94.0	1.1	4.0	**0.4**
10:50	90.0	X	10:51	94.0	1.0	4.0	**0.4**
10:51	90.0	X	10:52	94.0	1.0	4.0	**0.4**
10:52	90.0	X	10:53	94.0	1.0	4.0	**0.4**

See Next Page

[1]Silty Sand comprised of completely weathered friable decomposed granite
*Depth below reference datum
**Corrected for full depth gravel in annulus

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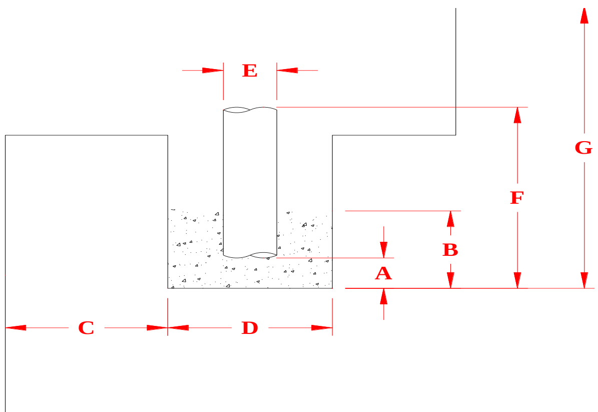
PERCOLATION TEST DATA SHEET

Project Name:	Glacier Point	Project No.:	160577
Project Location:	Yosemite Valley, CA	Pit No.:	P-7 pg 2 of 2



A.	Gravel Layer Depth, in.	2
B.	Total Gravel Thickness, in.	12
C.	Distance from Shelf, ft.	-
D.	Hole Diameter, in.	4.5 - 6.0
E.	Casing Diameter, in.	2
F.	Reference Depth, in.	98
G	Hole Depth, ft. (in.)	5 (60)
	Depth to Groundwater	-
Soil Type	Silty SAND (SM)[1]	

Date & Time Saturated 10/29/15 10:30 AM
Depth of Water after 24-hour Saturation 0

Begin Test	Initial Depth to Water*, in.	Refilled	End Test	Final Depth to Water*, in.	Test Duration, min.	Water Drop, in.	**Drop Rate min./in.****
10:53	90.0	X	10:54	94.0	1.0	4.0	**0.4**
10:54	90.0	X	10:55	94.0	1.0	4.0	**0.4**
10:55	90.0	X	10:56	94.0	1.0	4.0	**0.4**
10:56	90.0	X	10:57	94.0	1.0	4.0	**0.4**
10:57	90.0	X	10:58	94.0	1.0	4.0	**0.4**
10:58	90.0	X	10:59	94.0	1.0	4.0	**0.4**
10:59	90.0	X	11:00	94.0	1.0	4.0	**0.4**
11:00	90.0	X	11:01	94.0	1.0	4.0	**0.4**
11:01	90.0	X	11:02	94.0	1.0	4.0	**0.4**
11:02	90.0	X	11:03	94.0	1.0	4.0	**0.4**
11:03	90.0	X	11:04	94.0	1.0	4.0	**0.4**
11:04	90.0	X	11:05	94.0	1.0	4.0	**0.4**
11:05	90.0	X	11:06	94.0	1.0	4.0	**0.4**
11:06	90.0	X	11:07	94.0	1.0	4.0	**0.4**
11:07	90.0	X	11:08	94.0	1.0	4.0	**0.4**
11:08	90.0	X	11:09	94.0	1.0	4.0	**0.4**
11:09	90.0	X	11:10	94.0	1.0	4.0	**0.4**
11:10	90.0	X	11:11	94.0	1.0	4.0	**0.4**
11:11	90.0	X	11:12	94.0	1.0	4.0	**0.4**
11:12	90.0	X	11:13	94.0	1.0	4.0	**0.4**
11:13	90.0	X	11:14	94.0	1.0	4.0	**0.4**
11:14	90.0	X	11:15	94.0	1.0	4.0	**0.4**
11:15	90.0	X	11:16	94.0	1.0	4.0	**0.4**

Average (Last 6) = 0.4

[1]Silty Sand comprised of completely weathered friable decomposed granite
*Depth below reference datum
**Corrected for full depth gravel in annulus

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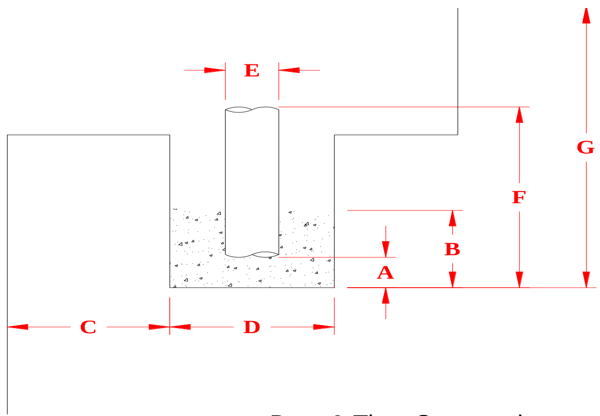
PERCOLATION TEST DATA SHEET

Project Name:	Glacier Point	Project No.:	160577
Project Location:	Yosemite Valley, CA	Pit No.:	P-8 pg 1 of 2



A.	Gravel Layer Depth, in.	2.5
B.	Total Gravel Thickness, in.	12
C.	Distance from Shelf, ft.	-
D.	Hole Diameter, in.	4.5 - 6.0
E.	Casing Diameter, in.	2
F.	Reference Depth, in.	64
G	Hole Depth, ft. (in.)	5 (60)
	Depth to Groundwater	-
Soil Type	Silty SAND (SM)[1]	

Date & Time Saturated 10/29/15 10:30 AM
Depth of Water after 24-hour Saturation 0

Begin Test	Initial Depth to Water*, in.	Refilled	End Test	Final Depth to Water*, in.	Test Duration, min.	Water Drop, in.	**Drop Rate min./in.****
10:30	56.0	X	10:31	60.0	1.0	4.0	**0.4**
10:31	56.0	X	10:32	60.0	1.1	4.0	**0.4**
10:32	56.0	X	10:33	60.0	1.2	4.0	**0.5**
10:33	56.0	X	10:34	60.0	1.2	4.0	**0.5**
10:34	56.0	X	10:35	60.0	1.2	4.0	**0.5**
10:35	56.0	X	10:37	60.0	1.2	4.0	**0.5**
10:37	56.0	X	10:38	60.0	1.2	4.0	**0.5**
10:38	56.0	X	10:39	60.0	1.2	4.0	**0.5**
10:39	56.0	X	10:40	60.0	1.2	4.0	**0.5**
10:40	56.0	X	10:42	60.0	1.2	4.0	**0.5**
10:42	56.0	X	10:43	60.0	1.2	4.0	**0.5**
10:43	56.0	X	10:44	60.0	1.3	4.0	**0.5**
10:44	56.0	X	10:46	60.0	1.3	4.0	**0.5**
10:46	56.0	X	10:47	60.0	1.3	4.0	**0.5**
10:47	56.0	X	10:49	60.0	1.2	4.0	**0.5**
10:49	56.0	X	10:50	60.0	1.3	4.0	**0.5**
10:50	56.0	X	10:51	60.0	1.3	4.0	**0.5**
10:51	56.0	X	10:53	60.0	1.3	4.0	**0.5**
10:53	56.0	X	10:54	60.0	1.3	4.0	**0.5**
10:54	56.0	X	10:55	60.0	1.3	4.0	**0.5**
10:55	56.0	X	10:57	60.0	1.3	4.0	**0.5**
10:57	56.0	X	10:58	60.0	1.3	4.0	**0.5**
10:58	56.0	X	10:59	60.0	1.3	4.0	**0.5**

See Next Page

[1]Silty Sand comprised of completely weathered friable decomposed granite
*Depth below reference datum
**Corrected for full depth gravel in annulus

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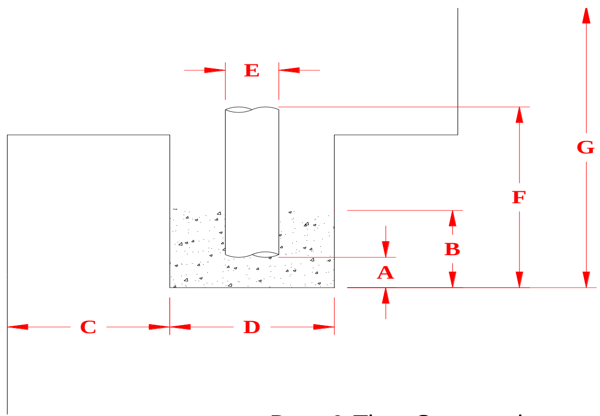
PERCOLATION TEST DATA SHEET

Project Name:	Glacier Point	Project No.:	160577
Project Location:	Yosemite Valley, CA	Pit No.:	P-8 pg 2 of 2



A.	Gravel Layer Depth, in.	2.5
B.	Total Gravel Thickness, in.	12
C.	Distance from Shelf, ft.	-
D.	Hole Diameter, in.	4.5 - 6.0
E.	Casing Diameter, in.	2
F.	Reference Depth, in.	64
G	Hole Depth, ft. (in.)	5 (60)
	Depth to Groundwater	-
Soil Type	Silty SAND (SM)[1]	

Date & Time Saturated 10/29/15 10:30 AM
Depth of Water after 24-hour Saturation 0

Begin Test	Initial Depth to Water*, in.	Refilled	End Test	Final Depth to Water*, in.	Test Duration, min.	Water Drop, in.	**Drop Rate min./in.****
10:59	56.0	X	11:01	60.0	1.3	4.0	**0.5**
11:01	56.0	X	11:02	60.0	1.3	4.0	**0.5**
11:02	56.0	X	11:03	60.0	1.3	4.0	**0.5**
11:03	56.0	X	11:05	60.0	1.3	4.0	**0.5**
11:05	56.0	X	11:06	60.0	1.3	4.0	**0.5**
11:06	56.0	X	11:07	60.0	1.3	4.0	**0.5**
11:07	56.0	X	11:09	60.0	1.3	4.0	**0.5**
11:09	56.0	X	11:10	60.0	1.3	4.0	**0.5**
11:10	56.0	X	11:11	60.0	1.3	4.0	**0.5**
11:11	56.0	X	11:13	60.0	1.3	4.0	**0.5**
11:13	56.0	X	11:14	60.0	1.3	4.0	**0.5**
11:14	56.0	X	11:15	60.0	1.3	4.0	**0.5**
11:15	56.0	X	11:17	60.0	1.3	4.0	**0.5**
11:17	56.0	X	11:18	60.0	1.3	4.0	**0.5**
11:18	56.0	X	11:19	60.0	1.3	4.0	**0.5**
11:19	56.0	X	11:21	60.0	1.3	4.0	**0.5**
11:21	56.0	X	11:22	60.0	1.3	4.0	**0.5**
11:22	56.0	X	11:23	60.0	1.3	4.0	**0.5**
11:23	56.0	X	11:25	60.0	1.3	4.0	**0.5**
11:25	56.0	X	11:26	60.0	1.3	4.0	**0.5**
11:26	56.0	X	11:27	60.0	1.3	4.0	**0.5**
11:27	56.0	X	11:29	60.0	1.3	4.0	**0.5**
11:29	56.0	X	11:30	60.0	1.3	4.0	**0.5**
						Average (Last 6) =	**0.5**

[1]Silty Sand comprised of completely weathered friable decomposed granite
*Depth below reference datum
**Corrected for full depth gravel in annulus

APPENDIX C

United States Department of the Interior

NATIONAL PARK SERVICE
Yosemite National Park
P. O. Box 577
Yosemite, California 95389

IN REPLY REFER TO:
L7615(YOSE-PM)

APR 06 2018

Memorandum

To: Garrett Chun, Project Manager, Yosemite National Park

From: Superintendent, Yosemite National Park

Subject: NEPA and NHPA Clearance: 2017-042 Glacier Point Leach Field Installation (77058)

The Superintendent and park interdisciplinary team have reviewed the proposed project and completed an impact analysis and documentation, and have determined the following:

- There will not be any effect on threatened, endangered, or rare species and/or their critical habitat.
- There will be no adverse effect to historical properties.
- There will not be serious or long-term undesirable environmental or visual effects.

The subject proposed project, therefore, is now cleared for all NEPA and NHPA compliance requirements as presented above. Project plans and specifications are approved and construction and/or project implementation can commence.

For the proposed project actions to be within compliance requirements during construction and/or project implementation, the following mitigations must be adhered to:

- After topsoil replacement, bare soils will be covered either with organic litter and duff mulch or with erosion control matting. Erosion control matting will consist of natural materials only, no biodegradable plastics. Notify vegetation staff 2-4 weeks in advance of topsoil replacement to arrange for seeding prior to mulching or matting.
- Before disturbing new leach field location, trenches, and staging area, contractor will first salvage organic litter and duff, then top 6 to 8 inches of topsoil and store these two materials separately for replacement after finish grading.
- Work limits and trees to remain should be marked and closely protected.
- Import fill will come from an approved, weed-free source- -consult with Invasive Plant Botanist. (Note that intermediate fill storage at Badger Pass or Chinquapin makes this important). Import fill will not be used within 6 inches of the surface. Trenches will be capped with at least 6 inches of native local soil.
- Equipment will be cleaned and inspected prior to entry into the park. Arrange equipment inspections with vegetation staff (Victor Goldman) 1-2 weeks in advance.
- Survey for sensitive and rare plant species in 2018, for avoidance or possible replanting following construction.

- Active revegetation (seeding with native herbaceous plants) is required for establishing native vegetation and preventing invasive plant dominance in the sunny, open conditions when the tree overstory is removed.

Recommendations for Conditions or Stipulations: None

For complete compliance information see PEPC Project 77058.

Michael T. Reynolds, Superintendent

Enclosure (with attachments)

cc: Statutory Compliance File

National Park Service
U.S. Department of the Interior

Yosemite National Park
Date: 03/22/2018

Categorical Exclusion Form

Project: 2017-042 Glacier Point Leach Field Installation
PEPC Project Number: 77058
Project Description:

This project will install a new sewer effluent disposal system to service the Glacier Point area consisting of the Glacier Point flush restrooms, quarter's cabins, and backwash generated from the water treatment system. This system includes a new disposal field, new dosing tank, relocation of potable water line, pump replacement, and utility dirt road realignment. The new system will connect to the quarter's cabin sewer and will consist of a new septic tank, small pump station, and adding to the force main. Project will remove 21 red firs and seven boulders. Project equipment needed includes excavators, backhoes, loaders, dozers, and dump trucks.

Project dimensions:

- Leachfield measures 4100 feet long by 2 feet wide and 2.5 feet deep
- Quarters 4 inch sewer line measures 440 feet long, 2 feet wide and 3.5 feet deep
- Force main (quarters to the dosing tank) measures 480 feet long, 2 feet wide and 2 feet deep
- Septic tank (2000 gallon) measures 11 feet long by 6 feet wide and 7 feet deep
- Small pump station measures 3 feet diameter by 11 feet deep
- Equalization tank (5750 gallon) measures 17 feet long by 8 wide and 8 feet deep
- Water line (2 inch) measures 600 feet long by 2 feet wide and 2.5 feet deep
- Utility Dirt Road Realignment measures 320 feet long by 12 feet wide and 0.5 feet deep

Glacier Point Road will have to be shared between the visitors and contractor. Due to the tight roads and life/safety concerns the project will utilize an area at least 75 feet and 125 feet for an intermediate staging site. Larger trucks can dump their supplies there and smaller trucks can take it through the narrow and windy road sections to the project site. Intermediate staging will be either the area at the Yosemite Ski & Snowboard parking lot (Site A) or at the parking lot next to the Chinquapin Restrooms (Site B).

Project Locations:

Mariposa County, CA

Mitigations:

- After topsoil replacement, bare soils will be covered either with organic litter and duff mulch or with erosion control matting. Erosion control matting will consist of natural materials only, no biodegradable plastics. Notify vegetation staff 2-4 weeks in advance of topsoil replacement to arrange for seeding prior to mulching or matting.
- Before disturbing new leach field location, trenches, and staging area, contractor will first salvage organic litter and duff, then top 6 to 8 inches of topsoil and store these two materials separately for replacement after finish grading.
- Work limits and trees to remain should be marked and closely protected.
- Import fill will come from an approved, weed-free source- -consult with Invasive Plant Botanist. (Note that intermediate fill storage at Badger Pass or Chinquapin makes this important). Import fill will not be used within 6 inches of the surface. Trenches will be capped with at least 6 inches of native local soil.

- Equipment will be cleaned and inspected prior to entry into the park. Arrange equipment inspections with vegetation staff (Victor Goldman) 1-2 weeks in advance.
- Survey for sensitive and rare plant species in 2018, for avoidance or possible replanting following construction.
- Active revegetation (seeding with native herbaceous plants) is required for establishing native vegetation and preventing invasive plant dominance in the sunny, open conditions when the tree overstory is removed.

CE Citation: C.15 Installation of underground utilities in previously disturbed areas having stable soils, or in an existing utility right-of-way.

Decision: I find that the action fits within the categorical exclusion above. Therefore, I am categorically excluding the described project from further NEPA analysis. No extraordinary circumstances apply.

Superintendent: Michael T. Reynolds ______ **Date:** 4/3/18

Michael T. Reynolds

Extraordinary Circumstances:

If implemented, would the proposal...	Yes/No	Notes
A. Have significant impacts on public health or safety?	No	
B. Have significant impacts on such natural resources and unique geographic characteristics as historic or cultural resources; park, recreation, or refuge lands; wilderness areas; wild or scenic rivers; national natural landmarks; sole or principal drinking water aquifers; prime farmlands; wetlands (Executive Order 11990); floodplains (Executive Order 11988); national monuments; migratory birds; and other ecologically significant or critical areas?	No	
C. Have highly controversial environmental effects or involve unresolved conflicts concerning alternative uses of available resources (NEPA section 102(2)(E))?	No	
D. Have highly uncertain and potentially significant environmental effects or involve unique or unknown environmental risks?	No	
E. Establish a precedent for future action or represent a decision in principle about future actions with potentially significant environmental effects?	No	
F. Have a direct relationship to other actions with individually insignificant, but cumulatively significant, environmental effects?	No	
G. Have significant impacts on properties listed or eligible for listing on the National Register of Historic Places, as determined by either the bureau or office?	No	
H. Have significant impacts on species listed or proposed to be listed on the List of Endangered or Threatened Species, or have significant impacts on designated Critical Habitat for these species?	No	
I. Violate a federal, state, local or tribal law or requirement imposed for the protection of the environment?	No	
J. Have a disproportionately high and adverse effect on low income or minority populations (EO 12898)?	No	
K. Limit access to and ceremonial use of Indian sacred sites on federal lands by Indian religious practitioners or adversely affect the physical integrity of such sacred sites (EO 130007)?	No	
L. Contribute to the introduction, continued existence, or spread of noxious weeds or non-native invasive species known to occur in the area or actions that may promote the introduction, growth, or expansion of the range of such species (Federal Noxious Weed Control Act and Executive Order 13112)?	No	

ENVIRONMENTAL SCREENING FORM (ESF)

Updated Sept 2015 per NPS NEPA Handbook

A. PROJECT INFORMATION

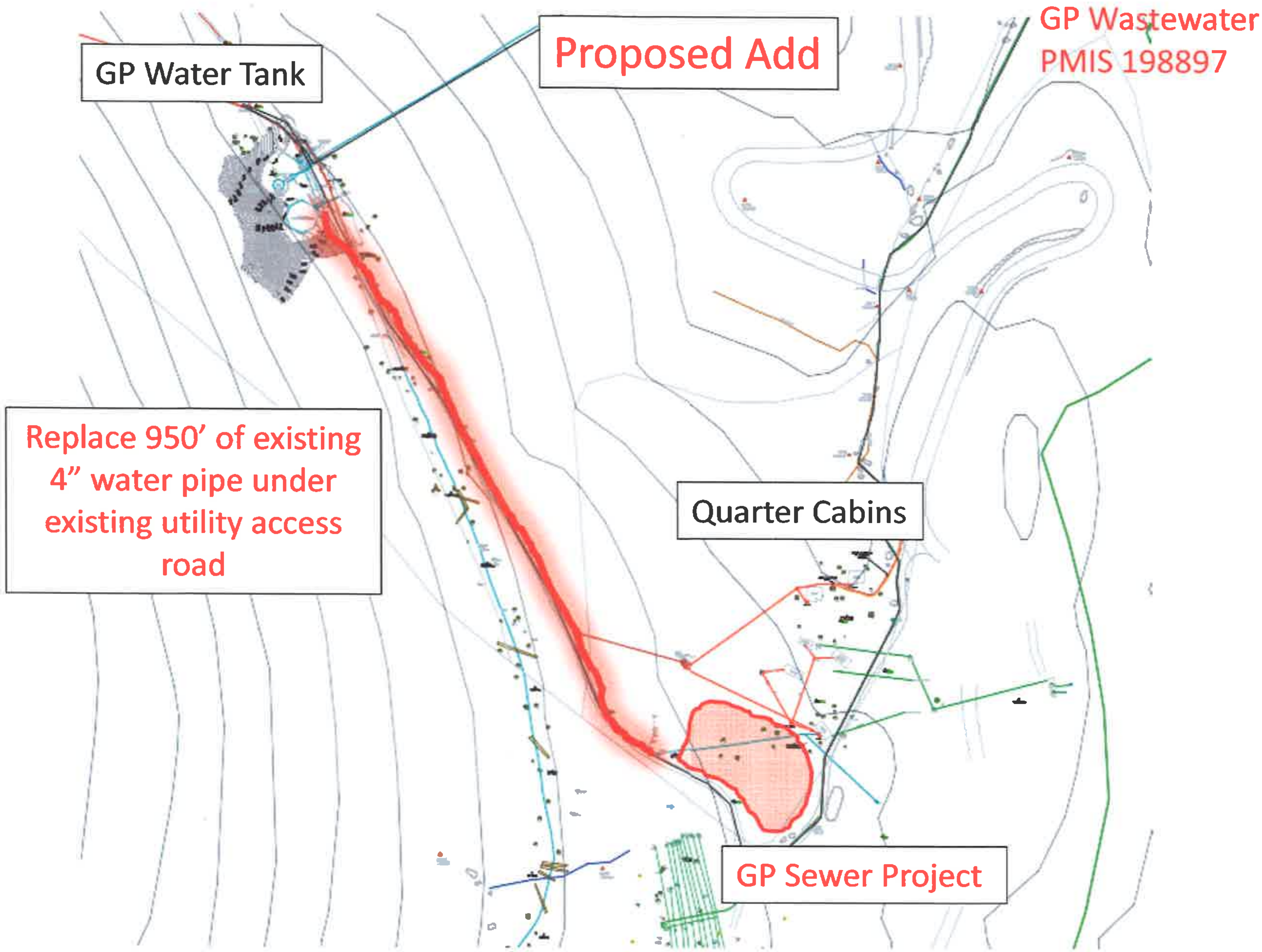
Project Title: 2017-042 Glacier Point Leach Field Installation
PEPC Project Number: 77058
Project Type: Capital Improvement/New Construction (CI)
Project Location:
County, State: Mariposa, California
Project Leader: Garrett Chun

B. RESOURCE IMPACTS TO CONSIDER:

Resource	Potential for Impact	Potential Issues & Impacts
Air Air Quality	Potential	This project will generate temporary dust and air emissions during the leach field installation.
Biological Nonnative or Exotic Species	Potential	Equipment will be cleaned and inspected prior to entry into the park. Imported fill will come from an approved, weed-free source. Imported fill will not be used within six inches of the surface.
Biological Species of Special Concern or Their Habitat	Potential	Sensitive and rare plant species survey will occur prior to work. Active re-vegetation (seeding with native herbaceous plants) is required for establishing native vegetation and preventing invasive plant dominance in the sunny, open conditions when the tree overstory is removed.
Biological Vegetation	Potential	Vegetation staff is assured that the impact of the 30+ large (and presumably several hundred year-old) red fir trees to be removed is offset by the substantial improvement to visitor experience and operational efficiency obtained by a functional leach field, and that there is high confidence, based on monitoring of groundwater wells, that this new leach field will be functional (sufficiently above the water table) for the desired length of the season.
Biological Wildlife and/or Wildlife Habitat including terrestrial and aquatic species	None	
Cultural Archeological Resources	None	

Cultural Cultural Landscapes	None	
Cultural Ethnographic Resources	None	This project went on the January 2018 tribal spreadsheet; no comments were received.
Cultural Museum Collections	None	
Cultural Prehistoric/historic structures	None	No historic buildings or structures are present therefore no historical architect review is required.
Geological Geologic Features	Potential	Before disturbing new leach field location, trenches, and staging area, the contractor will first salvage organic litter and duff, then top six to eight inches of topsoil and store these two materials separately for replacement after finish grading. After topsoil replacement, bare soils will be covered either with organic litter and duff mulch or with erosion control matting. Erosion control matting will consist of natural materials only, no biodegradable plastics. Notify vegetation staff 2-4 weeks in advance of topsoil replacement to arrange for seeding prior to mulching or matting.
Geological Geologic Processes	None	
Lightscapes Lightscapes	None	
Other Human Health and Safety	None	
Other Operational	None	
Socioeconomic Land Use	None	
Socioeconomic Minority and low-income populations, size, migration patterns, etc.	None	
Socioeconomic Socioeconomic	None	
Soundscapes Soundscapes	None	
Viewsheds Viewsheds	Potential	Construction of the leach field will result in temporary equipment noises.
Visitor Use and	None	

Experience Recreation Resources		
Visitor Use and Experience Visitor Use and Experience	None	
Water Floodplains	None	
Water Marine or Estuarine Resources	None	
Water Water Quality or Quantity	None	
Water Wetlands	None	
Water Wild and Scenic River	None	
Wilderness Wilderness	None	

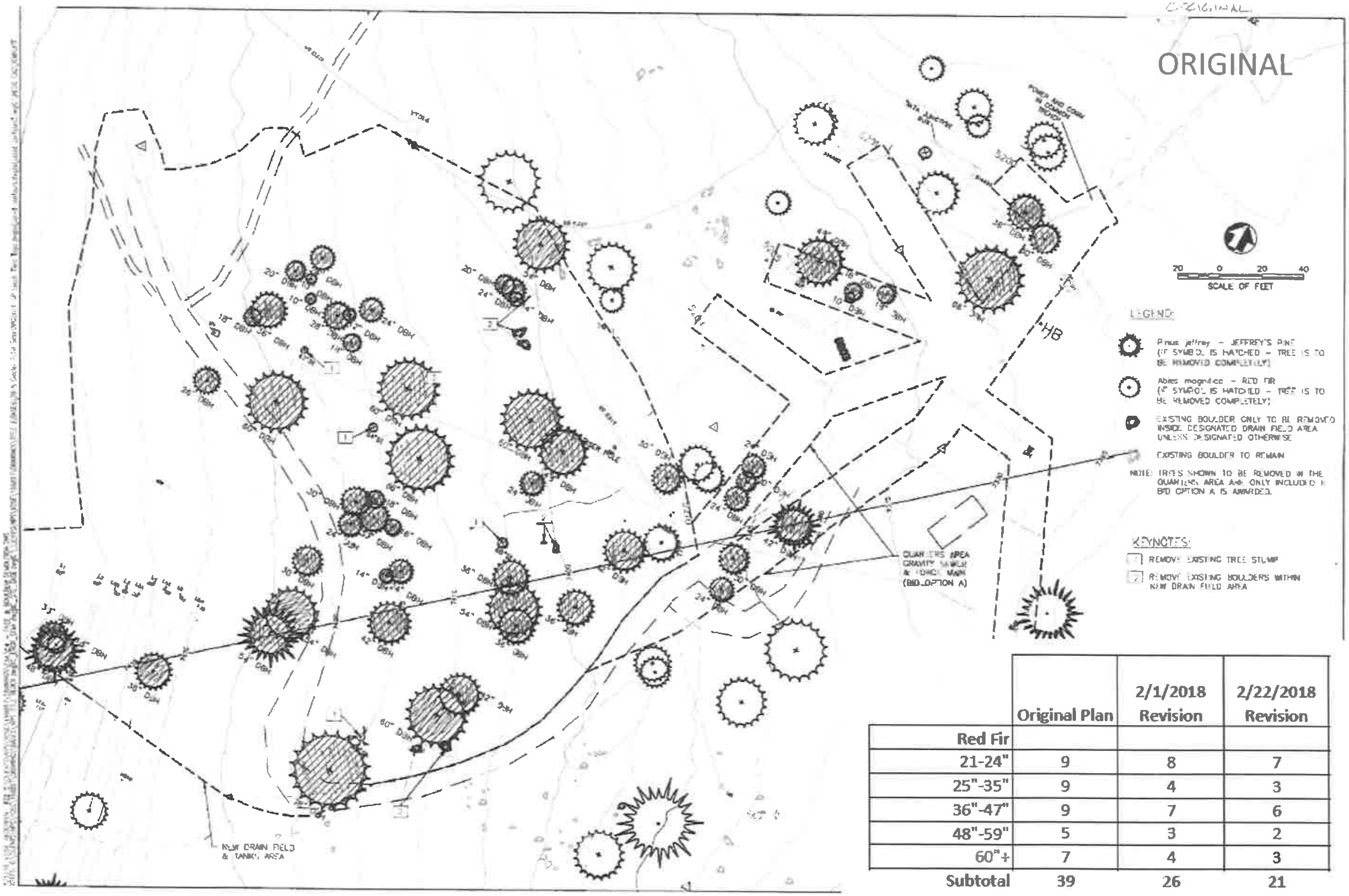

GP Wastewater
PMIS 198897
Proposed Add
GP Water Tank
Replace 950’ of existing 4” water pipe under existing utility access road
Quarter Cabins
GP Sewer Project

GLACIER POINT SEWER
PMIS 198897

TREE REMOVAL COUNT

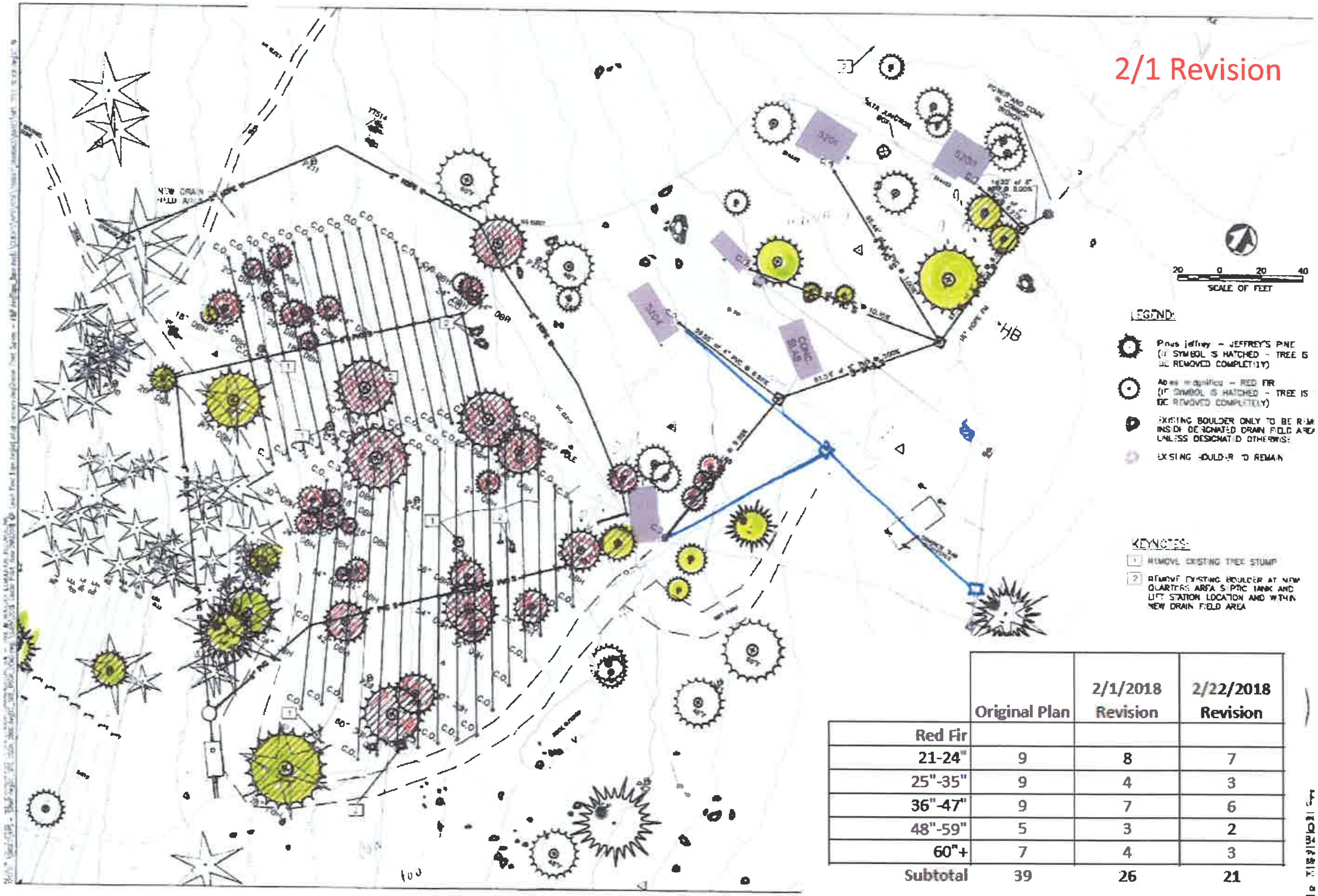
	Original Plan	2/1/2018 Revision	2/22/2018 Revision
Red Fir			
21-24"	9	8	7
25"-35"	9	4	3
36"-47"	9	7	6
48"-59"	5	3	2
60"+	7	4	3
Subtotal	**39**	**26**	**21**
Jeffrey Pine			
21-24"	0	0	0
25"-35"	0	0	0
36"-47"	1	0	0
48"-59"	2	0	0
60"+	0	0	0
Subtotal	**3**	**0**	**0**
TOTAL	**42**	**26**	**21**

Note: Trees 20" and under are not included above



	Original Plan	2/1/2018 Revision	2/22/2018 Revision
Red Fir			
21-24"	9	8	7
25"-35"	9	4	3
36"-47"	9	7	6
48"-59"	5	3	2
60"+	7	4	3
Subtotal	39	26	21

2/1 Revision



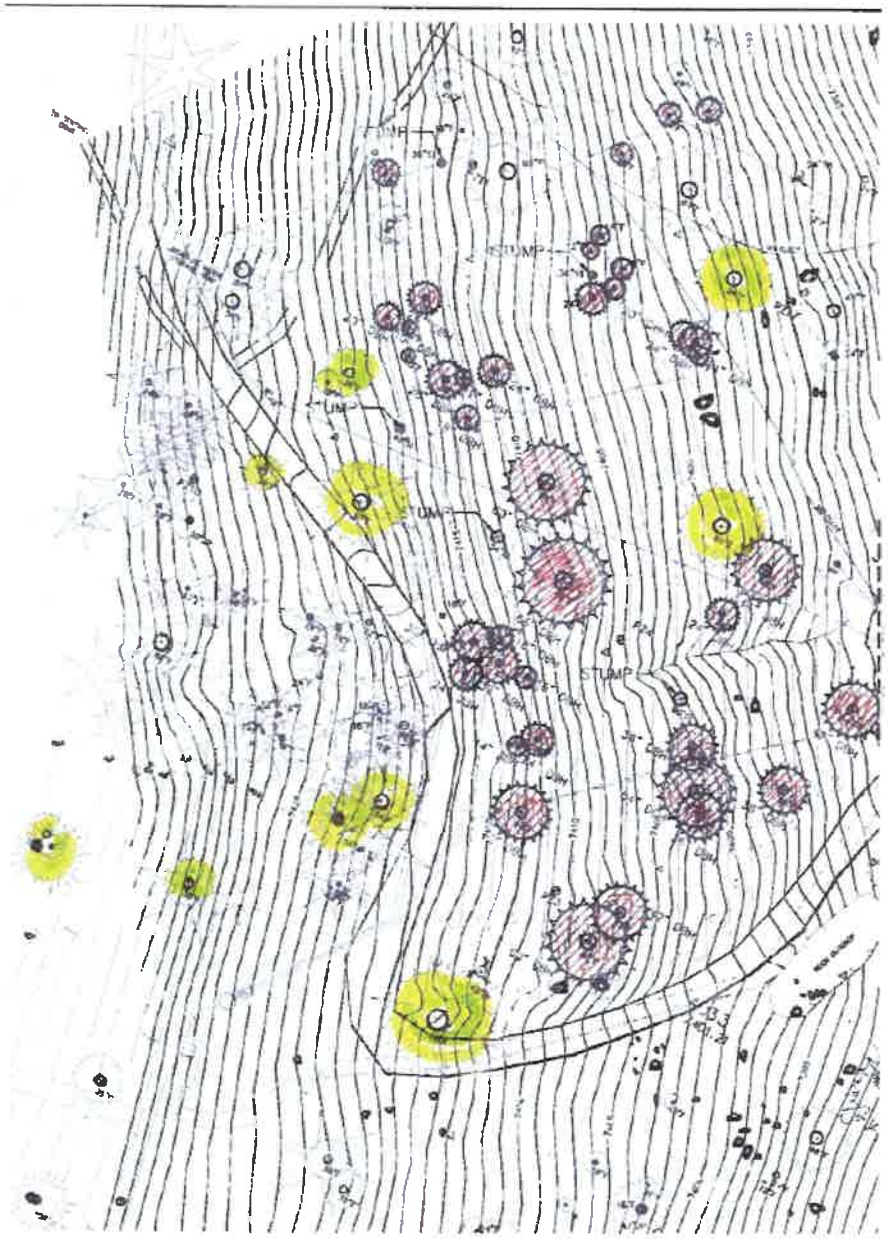
LEGEND:

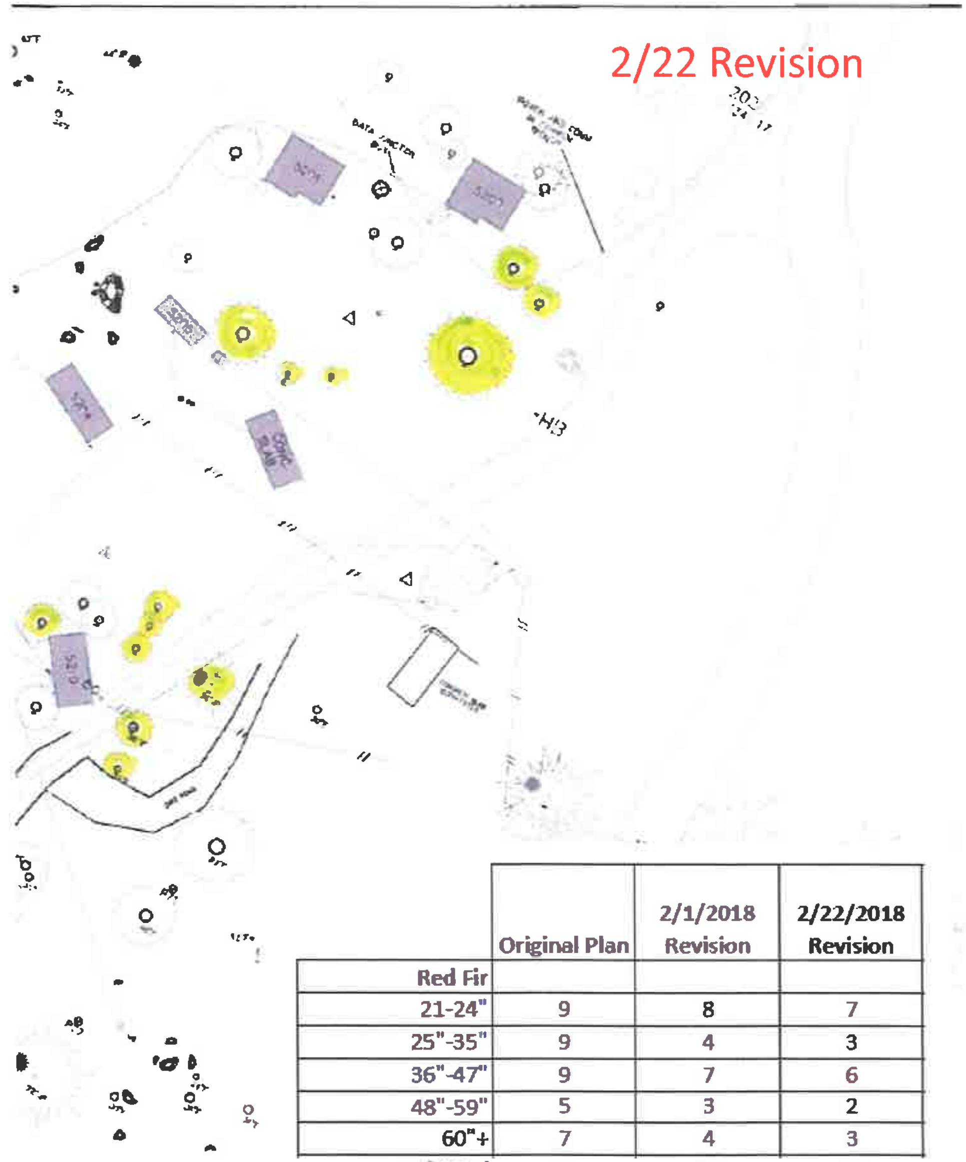
- Pinus jeffreyi – JEFFREY'S PINE (IF SYMBOL IS HATCHED – TREE IS TO BE REMOVED COMPLETELY)
- Abies magnifica – RED FIR (IF SYMBOL IS HATCHED – TREE IS TO BE REMOVED COMPLETELY)
- EXISTING BOULDER ONLY TO BE REMOVED INSIDE DESIGNATED DRAIN FIELD AREA UNLESS DESIGNATED OTHERWISE
- EXISTING BOULDER TO REMAIN

KEYNOTES:

1. REMOVE EXISTING TREE STUMP
2. REMOVE EXISTING BOULDER AT NEW QUARTERS AREA SEPTIC TANK AND LIFT STATION LOCATION AND WITHIN NEW DRAIN FIELD AREA

	Original Plan	2/1/2018 Revision	2/22/2018 Revision
Red Fir			
21-24"	9	8	7
25"-35"	9	4	3
36"-47"	9	7	6
48"-59"	5	3	2
60"+	7	4	3
Subtotal	39	26	21





2/22 Revision

	Original Plan	2/1/2018 Revision	2/22/2018 Revision
Red Fir			
21-24"	9	8	7
25"-35"	9	4	3
36"-47"	9	7	6
48"-59"	5	3	2
60"+	7	4	3
Subtotal	39	26	21

ASSESSMENT OF ACTIONS HAVING AN EFFECT ON HISTORIC PROPERTIES

A. DESCRIPTION OF UNDERTAKING

1. Park: Yosemite National Park

2. Project Description:

Project Name: 2017-042 Glacier Point Leach Field Installation
Prepared by: Renea Kennec **Date Prepared:** 01/23/2018 **Telephone:** 209-379-1038
PEPC Project Number: 77058
Locations:
County, State: Mariposa, CA

Area of potential effects (as defined in 36 CFR 800.16[d])
The project is located near Glacier Point. The existing septic and leach field system supports the Glacier Point restrooms. When completed, the project will also support the nearby Quarters Cabins. The drain field will be about three quarters of an acre. The old drain field will be abandoned in place. The majority of the project will not be visible and will be re-vegetated with grasses. The area is surrounded by trees however the above ground control panel may be visible from the adjacent housing to the north. The vertical APE for the leach and septic system will not exceed six feet in depth and would be specific to the leach field and force main and gravity lines. Excavation for the septic tanks, manholes and wet wells will not exceed ten feet in depth.

3. Has the area of potential effects been surveyed to identify historic properties?

___ **No**
X **Yes**
Source or reference:

4. Potentially Affected Resources:

Archeological Resources Affected: No

Historical Structures/Resources Affected: No
Historical Structures/Resources Notes: Project is adjacent to the Glacier Point Road Historic District and nearby residential buildings. Work will not affect historic properties.

Cultural Landscapes Affected: No

Ethnographic Resources Affected: No

5. The proposed action will: (check as many as apply)

No Destroy, remove, or alter features/elements from a historic structure
No Replace historic features/elements in kind

No Add non-historic features/elements to a historic structure

No Alter or remove features/elements of a historic setting or environment (inc. terrain)

No Add non-historic features/elements (inc. visual, audible, or atmospheric) to a historic setting or cultural landscape

No Disturb, destroy, or make archeological resources inaccessible

No Disturb, destroy, or make ethnographic resources inaccessible>

Yes Potentially affect presently unidentified cultural resources

No Begin or contribute to deterioration of historic features, terrain, setting, landscape elements, or archeological or ethnographic resources

No Involve a real property transaction (exchange, sale, or lease of land or structures)

____ Other (please specify): ________________________________

6. Supporting Study Data:
(Attach if feasible; if action is in a plan, EA or EIS, give name and project or page number.)

B. REVIEWS BY CULTURAL RESOURCE SPECIALISTS

The park 106 coordinator requested review by the park's cultural resource specialist/advisors as indicated by check-off boxes or as follows:

[X] 106 Advisor
Name: Kimball Koch
Date: 03/19/2018
Comments: Per Cultural Resources Program Manager, no historic buildings or structures are present therefore no historical architect review is required.

***Check if project does not involve ground disturbance* [ ]**
Assessment of Effect: __No Potential to Cause Effect __X__No Historic Properties Affected __No Adverse Effect __Adverse Effect __Streamlined Review
Recommendations for conditions or stipulations:

Doc Method: Standard 4-Step Process

[X] Anthropologist
Name: Scott Carpenter
Date: 03/19/2018
Comments: Project was on the January 2018 tribal consultation spreadsheet for 30-day review. Comment received from North Fork Rancheria. Park responded; response did not affect the project.

***Check if project does not involve ground disturbance* [ ]**
Assessment of Effect: __No Potential to Cause Effect __X__No Historic Properties Affected __No Adverse Effect __Adverse Effect __Streamlined Review
Recommendations for conditions or stipulations:

Doc Method: Standard 4-Step Process

[X] **Archeologist**
Name: Sara Dolan
Date: 03/19/2018
Comments: No archeological sites in the project area.

Check if project does not involve ground disturbance [ ]
Assessment of Effect: __No Potential to Cause Effect __X__No Historic Properties Affected __No Adverse Effect __Adverse Effect __Streamlined Review
Recommendations for conditions or stipulations:

Doc Method: Standard 4-Step Process

[X] **Historian**
Name: Scott Carpenter
Date: 03/19/2018
Comments: No historic buildings or structures are present therefore no historical architect review is required.

Check if project does not involve ground disturbance [ ]
Assessment of Effect: __No Potential to Cause Effect __X__No Historic Properties Affected __No Adverse Effect __Adverse Effect __Streamlined Review
Recommendations for conditions or stipulations:

Doc Method: Standard 4-Step Process

[X] **Historical Landscape Architect**
Name: Kimball Koch
Date: 03/07/2018

Check if project does not involve ground disturbance [ ]
Assessment of Effect: __No Potential to Cause Effect __X__No Historic Properties Affected __No Adverse Effect __Adverse Effect __Streamlined Review
Recommendations for conditions or stipulations:

Doc Method: Standard 4-Step Process

No Reviews From: Curator, Historical Architect, Other Advisor

---

C. PARK SECTION 106 COORDINATOR'S REVIEW AND RECOMMENDATIONS

1. Assessment of Effect:

______No Potential to Cause Effects
__X__No Historic Properties Affected
______No Adverse Effect
______Adverse Effect

2. Documentation Method:

[ ] A. Standard 36 CFR Part 800 Consultation
Further consultation under 36 CFR Part 800 is needed.

[X] B. Streamlined Review Under the 2008 Servicewide Programmatic Agreement (PA)
The above action meets all conditions for a streamlined review under section III of the 2008 Servicewide PA for Section 106 compliance.

Applicable Streamlined Preview Criteria
(Specify 1-16 of the list of streamlined review criteria.)

[ ] C. Undertaking Related to Another Agreement
The proposed undertaking is covered for Section 106 purposes under another document such as a statewide agreement established in accord with 36 CFR 800.7 or counterpart regulations.

[ ] D. Combined NEPA/NHPA Process
Documentation is required for the preparation of an EA/FONSI or an EIS/ROD has been developed and used so as also to meet the requirements of 36 CFR 800.3 through 800.6

[ ] E. Memo to Project File

3. Consultation Information

SHPO Required: No

THPO Required: Yes
THPO Sent: Jan 10, 2018
THPO Received: Comment received from North Fork Rancheria. Park responded; response did not affect the project.

SHPO/THPO Notes: January 2018 Tribal Spreadsheet; one comment received from Northfork Rancheria.

Advisory Council Participating: No
Additional Consulting Parties: No

4. Stipulations and Conditions: Following are listed any stipulations or conditions necessary to ensure that the assessment of effect above is consistent with 36 CFR Part 800 criteria of effect or to avoid or reduce potential adverse effects.

5. Mitigations/Treatment Measures: Measures to prevent or minimize loss or impairment of historic/prehistoric properties: (Remember that setting, location, and use may be relevant.)

No Assessment of Effect mitigations identified.

6. Assessment of Effect Notes:

D. RECOMMENDED BY PARK SECTION 106 COORDINATOR:

Kimball Koch _Kimball E Koch_ **Date:** 3/29/2018

E. SUPERINTENDENT'S APPROVAL

The proposed work conforms to the NPS *Management Policies* and *Cultural Resource Management Guideline*, and I have reviewed and approve the recommendations, stipulations, or conditions noted in Section C of this form.

Superintendent: _Michael Reynolds_ **Date:** 4/3/18